

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



December 27, 2006

MATERIALS LABORATORY FACTUAL REPORT

Report No. 06-105

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : I-90 connector tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Epoxy samples prepared in the Materials Laboratory and samples of the adhesive anchor material from the I-90 tunnel.

C. DETAILS OF THE EXAMINATION

The adhesive that was used in the anchorages in the I-90 tunnel roof had been preliminarily identified in construction documents as a Power-Fast epoxy¹. Power-Fast epoxy is sold in two formulations: Standard Set and Fast Set. Power-Fast product literature² reports that the Standard Set formulation takes about twice as long to gel (begin to harden) as the Fast Set formulation. As an example, at 75 °F, the gel time for the Standard Set epoxy is listed as 35 minutes while the gel time for the Fast Set epoxy is listed as 15 minutes. Two 22-ounce packages of each type of epoxy were purchased for testing, along with a manual injection gun. Each package of epoxy consists of two side-by-side tubes, one containing a white epoxy resin and the other containing a dark gray hardener. The injection gun simultaneously pushes two plungers that extrude the resin and hardener through a static mixing nozzle attached to both tubes. Each package of epoxy includes a static mixing nozzle. For both the Standard Set and the Fast Set, the resin tube and the hardener tube are of equal diameter, so the resin and hardener are mixed in a 1:1 ratio by volume. Both of the Standard Set packages had the same Lot Number of 06228OH1LC and Use By date of 08/2008. One Fast Set package had a Lot Number of 06194OI1LC and Use By date of 08/2008, and the other package had a Lot Number of 06208OI2LC and Use By date of 08/2008. The first 5 digits of the Lot Number indicate the year (06 = 2006) and the day of the year (228 = August 16; 194 = July 13; 208 = July 27)

¹ Powers Fasteners, 2 Powers Lane, Brewster, NY 10509

² Powers Fasteners Power-Fast® + Specification and Design Manual;
<http://www.powers.com/pdfs/chemical/08402.pdf>

when the materials were packaged. Samples discussed in this report were taken only from one package each of the Standard Set and Fast Set.

Samples of epoxy were first taken by extruding the resin and hardener through the static mixer nozzle to create samples mixed 1:1 by volume. These samples are identified as STD 50/50 or FAST 50/50 in the appended testing reports. In each case, the first nozzle-full of extruded epoxy was discarded to ensure that the material was mixed in the correct ratio. The static mixer nozzle was subsequently removed and separate quantities of the resin and hardener were extruded into containers. Resin and hardener from each epoxy formulation were then mixed at mass ratios of 70 percent hardener to 30 percent resin, 60 percent hardener to 40 percent resin, 40 percent hardener to 60 percent resin, and 30 percent hardener to 70 percent resin. The mixtures were allowed to cure for several days at which time most appeared to have fully hardened. However, the samples mixed at 70 percent hardener to 30 percent resin from both formulations did not harden completely, achieving a consistency of putty. Mixtures at the extreme ratios of 70 percent hardener to 30 percent resin (STD 70/30 or FAST 70/30 in the appended report) and 30 percent hardener to 70 percent resin (STD 30/70 or FAST 30/70 in the appended report) were submitted for testing with the 50/50 mixtures.

For both the Standard Set and Fast Set materials, samples of the resin, the hardener and the 1:1 mixed and cured epoxy were examined using X-ray energy dispersive spectroscopy (EDS) in the scanning electron microscope (SEM). The spectra from these samples are shown in figures 1 – 6. Differences among these spectra could be a result of the sample consistency; the resin and hardener samples were viscous fluids while the mixed and cured epoxy samples were solid. The spectra for the cured epoxy samples appear very similar in elemental content and relative peak height ratios, with peaks for carbon, oxygen, magnesium, silicon and calcium. Peaks for carbon and oxygen would be expected as elements of the epoxy network. The Material Safety Data Sheets for the Standard Set and Fast Set epoxies list calcium carbonate and talc (magnesium silicate) as components of both the resin and hardener.

Two samples of anchor adhesive material from the I-90 tunnel were also examined using EDS, and the resulting spectra are shown in figures 7 and 8. The anchorages involved in the accident were in the EB081-M1 row³. Sample EB081-M1-2/3/4-NW was powdery material that was dislodged from the hole during a borescope examination of the hole. Sample WB082-S-2-N was a fragment of the adhesive that was found loose in the bag containing the bolt from that location; the fragment retained a molded impression of the threads from the bolt. Spectra from both of these samples appeared similar in elemental make-up, with peaks for carbon, oxygen, magnesium, aluminum, silicon and calcium.

Samples of the epoxies mixed in the Materials Laboratory and samples of adhesive anchor material from the tunnel were also submitted to an outside laboratory⁴ for additional

³ The labels for the samples indicate the location of the anchor as to tunnel (EB = eastbound, WB = westbound), module (081 or 082 for these samples), row (M1 = middle1, S = south), hangers attached at that location (2/3/4 or just 2 for these samples) and bolt position (NW = northwest, N = north).

⁴ Trace Laboratories –East, 5 North Park Drive, Hunt Valley, MD

testing. The requested testing included Fourier Transform Infrared Spectroscopy (FTIR), Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA) and headspace Gas Chromatograph Mass Spectrometry (GCMS). Not all samples were tested with each method. Two reports documenting the results of that testing are appended; the samples tested are identified on the first page of each report⁵. Samples EB081-M1-2/3/4-NW, EB081-M2-2/3/4-SE and WB082-S-2-S represented the typical white fractured adhesive observed on the anchor bolts and in the anchor holes in the roof. Sample EB081-M1-2/3/4-NW was powdery material that was dislodged from the hole during a borescope examination of the hole. Samples EB081-M2-2/3/4-SE and WB082-S-2-S were chipped off of the anchor bolts from those locations. Sample WB083-M-9/10/11-SW was powdery material with a yellow tint that was scraped from the surface of the anchor bolt. Sample EB080-N-4/5/6-NE was a gummy brown material scraped from the surface of the anchor bolt.

Carl R. Schultheisz
Materials Research Engineer

⁵ Trace Laboratories reports S/O #59725 and S/O #60230

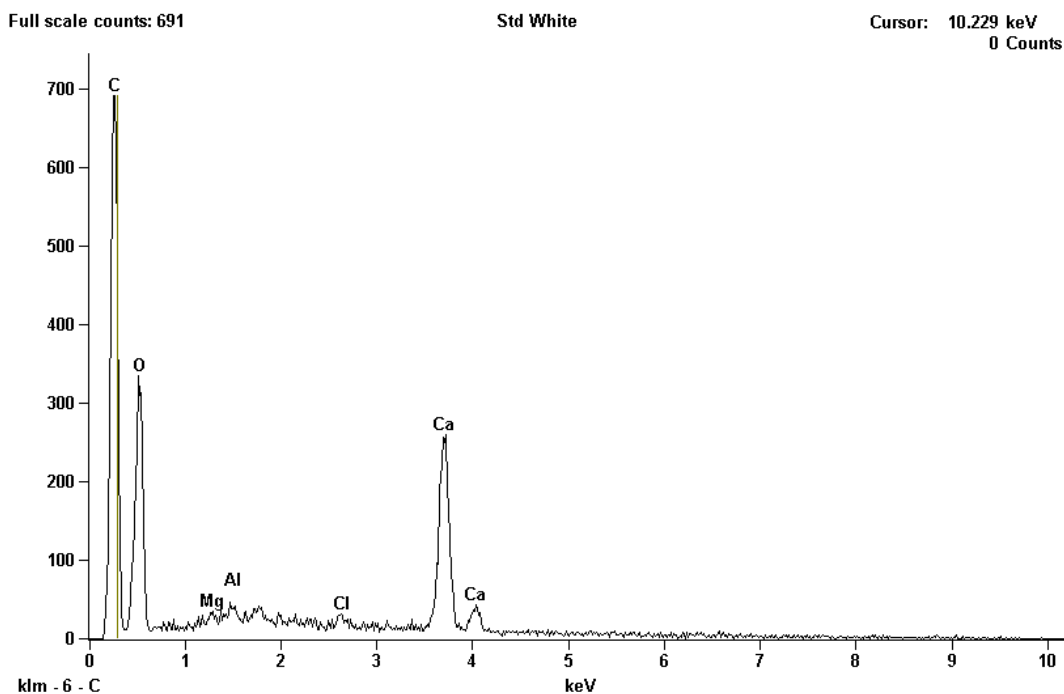


Figure 1. EDS spectrum of Standard Set epoxy resin, with peaks for carbon, oxygen, magnesium, aluminum, chlorine and calcium identified.

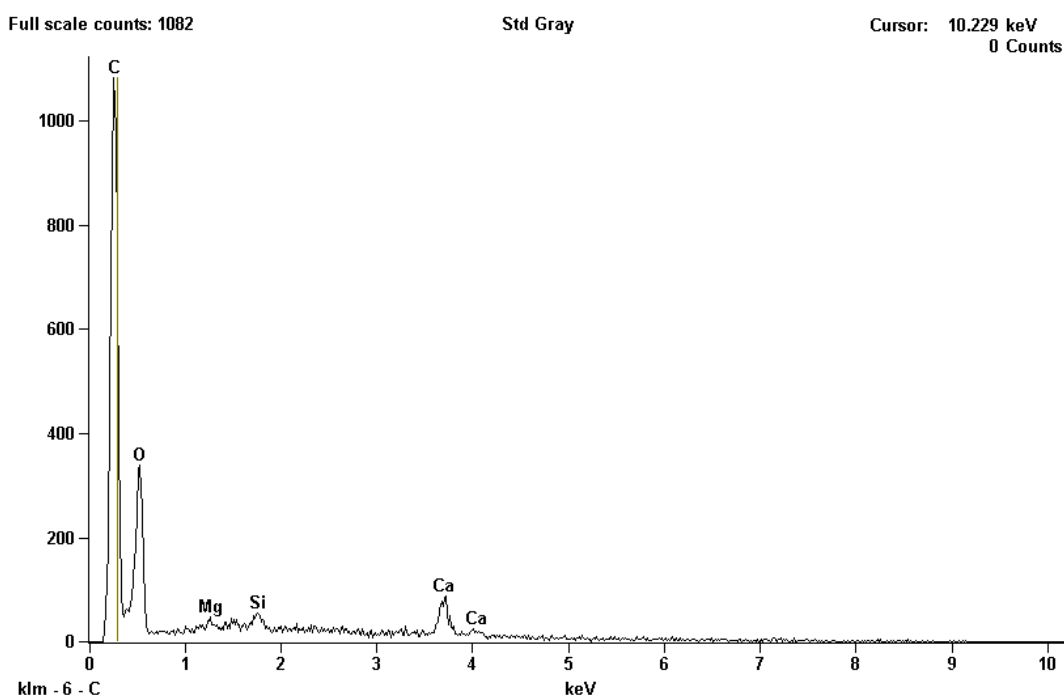


Figure 2. EDS spectrum of Standard Set hardener, with peaks for carbon, oxygen, magnesium, silicon and calcium identified.

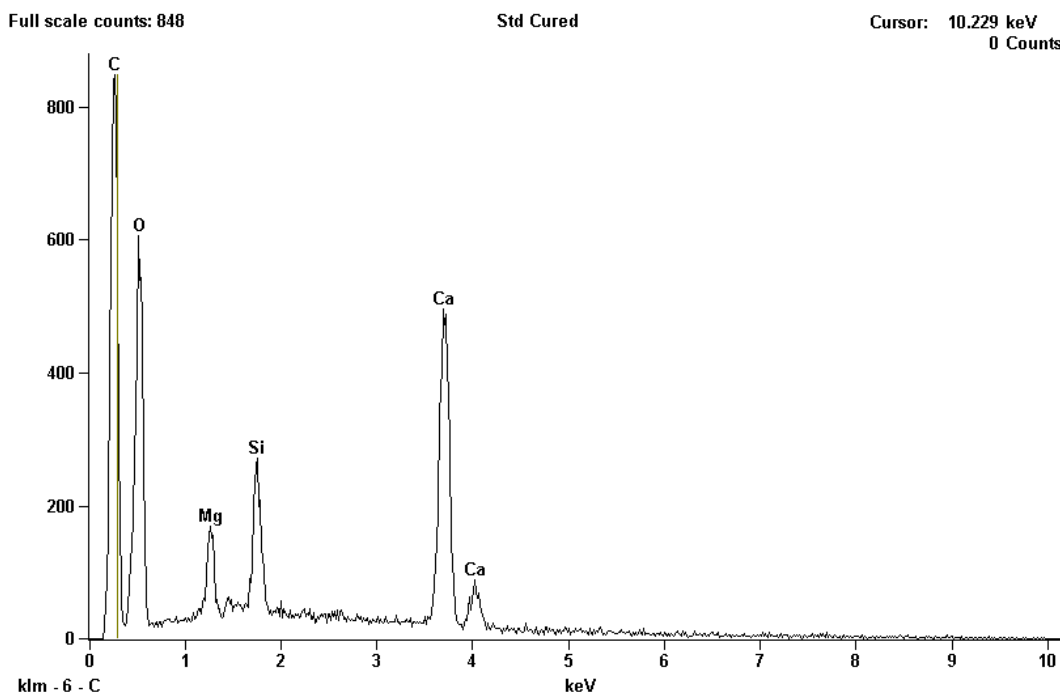


Figure 3. EDS spectrum of mixed and cured Standard Set epoxy, with peaks for carbon, oxygen, magnesium, silicon and calcium identified.

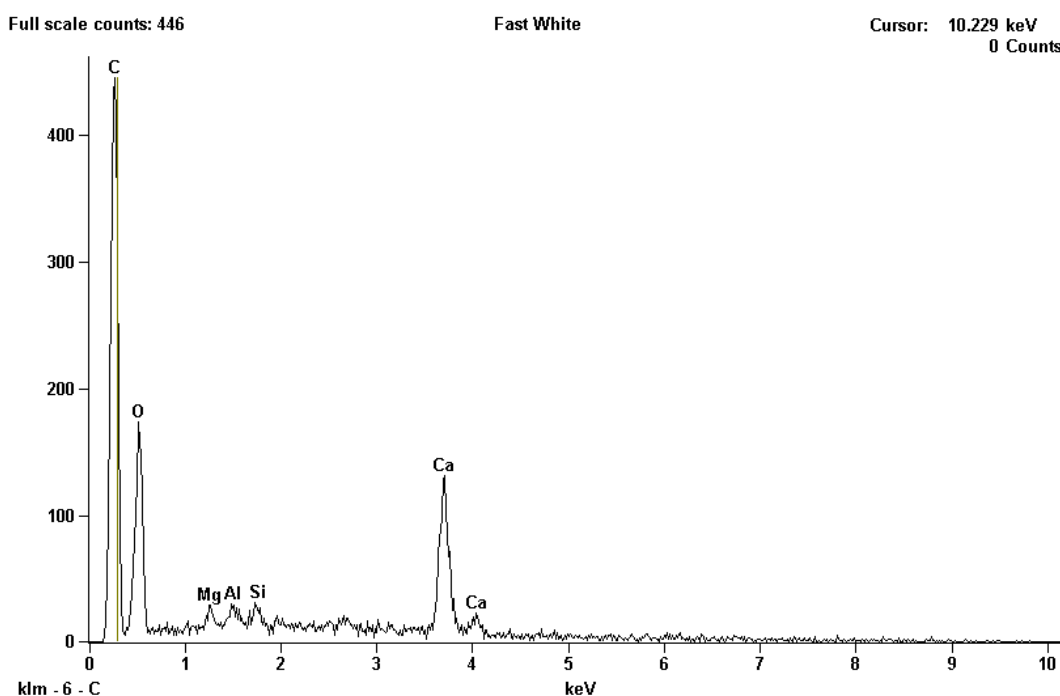


Figure 4. EDS spectrum of Fast Set epoxy resin, with peaks for carbon, oxygen, magnesium, aluminum, silicon and calcium identified.

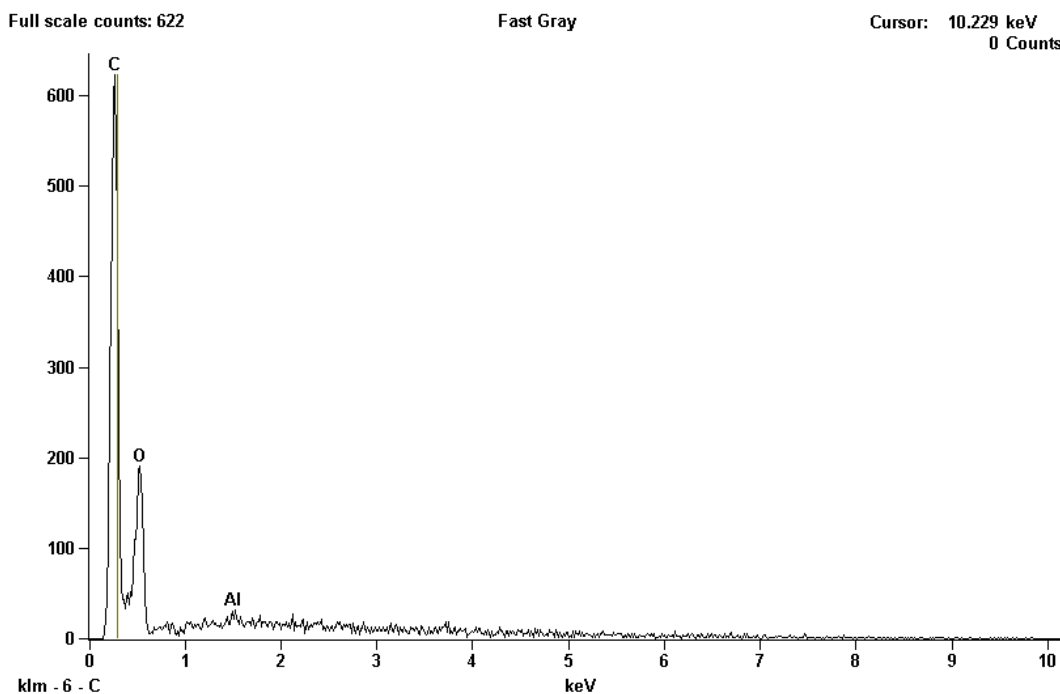


Figure 5. EDS spectrum of Fast Set hardener, with peaks for carbon, oxygen and aluminum identified.

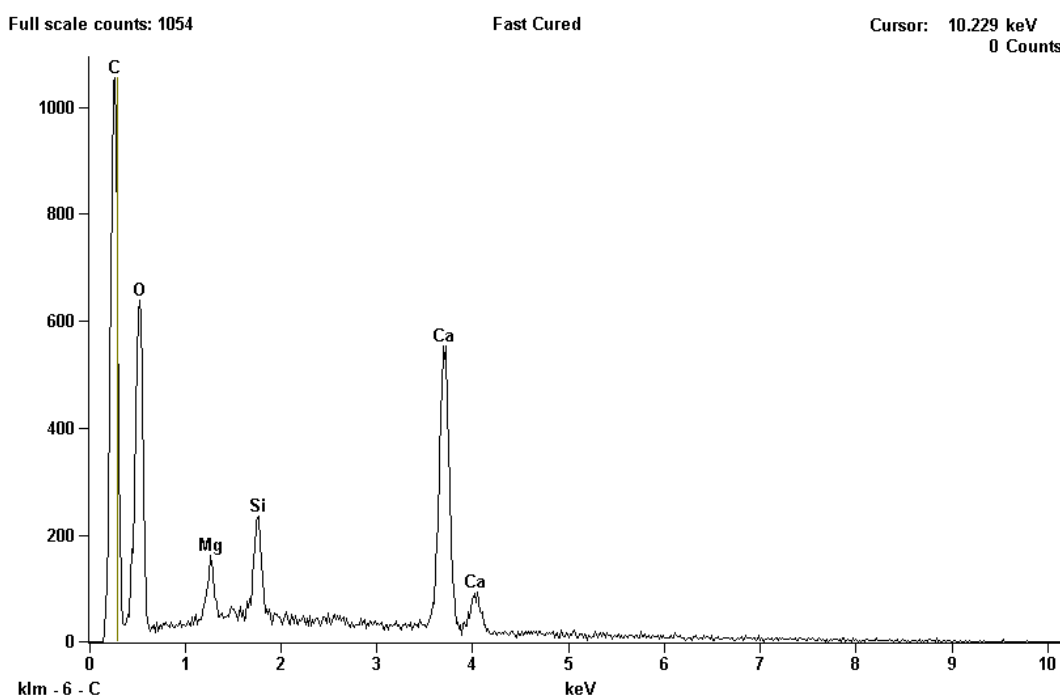


Figure 6. EDS spectrum of mixed and cured Fast Set epoxy, with peaks for carbon, oxygen, magnesium, silicon and calcium identified.

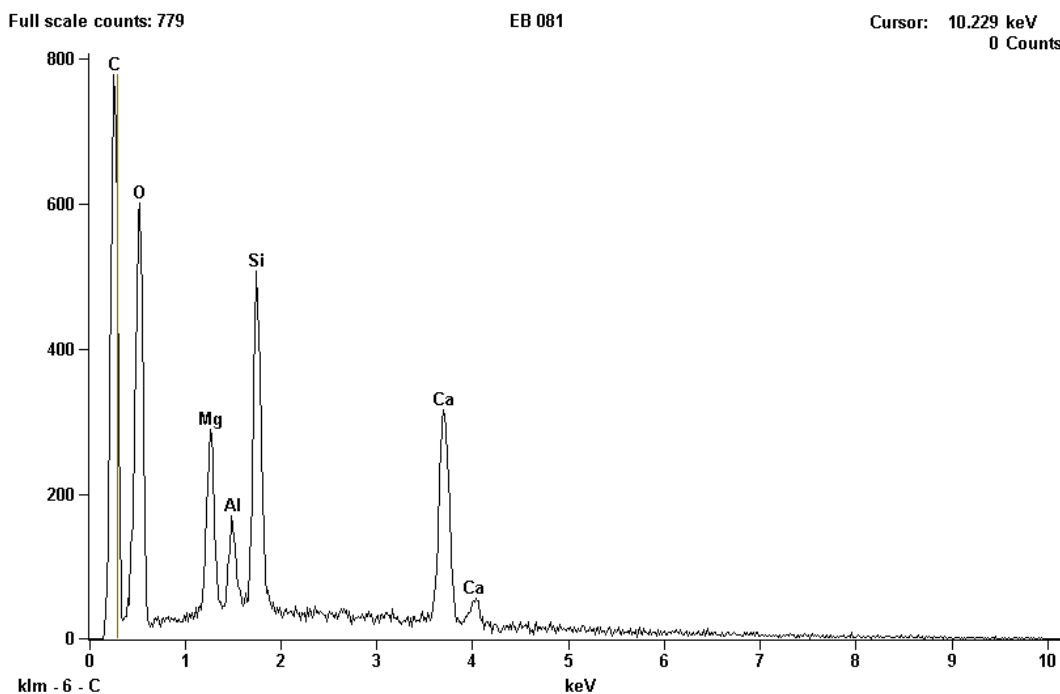


Figure 7. EDS spectrum of a sample of the adhesive taken from the hole at position EB081-M1-2/3/4-NW, with peaks for carbon, oxygen, magnesium, aluminum, silicon and calcium identified.

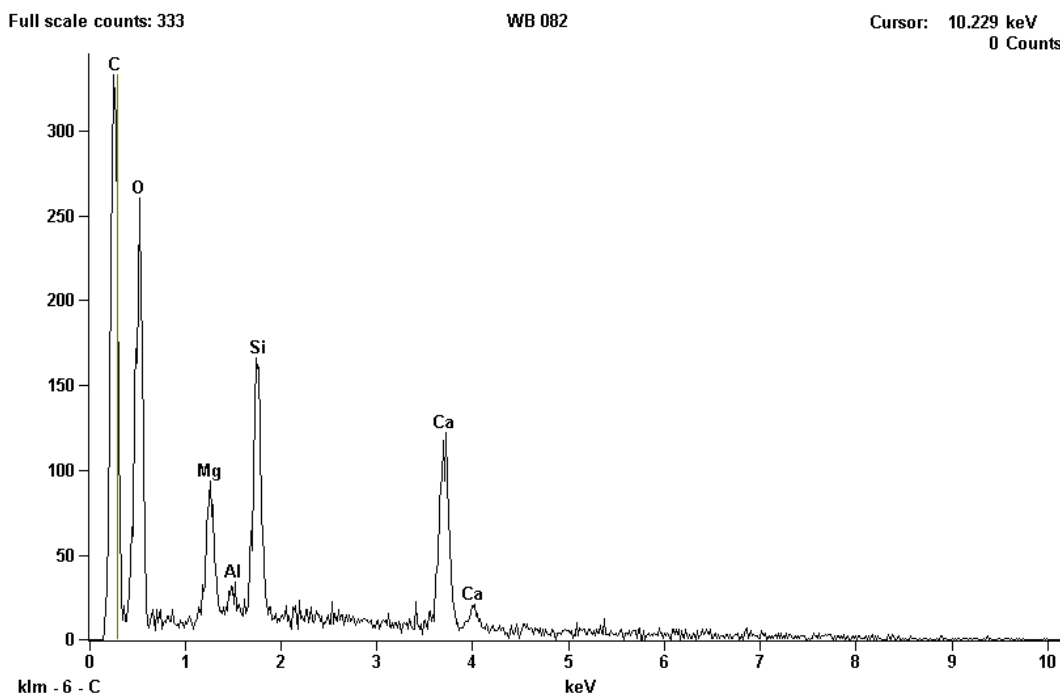


Figure 8. EDS spectrum of a sample of the adhesive taken from the anchor bolt at position WB082-S-2N, with peaks for carbon, oxygen, magnesium, aluminum, silicon and calcium identified.



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TEST REPORT FOR:

NTSB
RE-30 Materials Laboratory Division
490 L'Enfant Plaza East, S.W.
Washington, DC 20594

Attn: Carl Schultheisz

DATE IN:

September 12, 2006

P/O #:

NTSBP060084

SUBMISSION IDENTIFICATION:

Five (5) "unknown" and six (6) "known" epoxy samples were submitted for Fourier Transform Infrared Spectroscopy (FTIR) Analysis, Differential Scanning Calorimetry (DSC) Analysis, and Thermogravimetric Analyzer (TGA) Analysis to compare and contrast the samples from NTSB Accident Investigation # HWY06MH024.

The five (5) "unknown" and six (6) "known" samples were serialized by the customer as follows:

"Unknown" Samples

- EB081 M1 2/3/4 NW
- EB081 M2 2/3/4 SE
- WB082 S 2 S
- WB083 M 9/10/11 SW
- EB080 N 4/5/6 NE

"Known Samples"

- Std 50/50
- Std 70/30
- Std 30/70
- Fast 50/50
- Fast 70/30
- Fast 30/70

We offer our results and observations.

APPROVED:

John M. Radman
Senior Technical Director





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FOURIER TRANSFORM INFRARED SPECTROSCOPY (FTIR)

TEST SPECIMENS:

Five (5) “unknown” and six (6) “known” epoxy samples, identified as:

“Unknown” Samples

- EB081 M1 2/3/4 NW
- EB081 M2 2/3/4 SE
- WB082 S 2 S
- WB083 M 9/10/11 SW
- EB080 N 4/5/6 NE

“Known Samples”

- Std 50/50
- Std 70/30
- Std 30/70
- Fast 50/50
- Fast 70/30
- Fast 30/70

SPECIMEN PREPARATION:

A representative specimen from each submitted epoxy sample was removed using a clean stainless steel scalpel. The specimens were then individually mounted onto IR-reflective brass plates and rolled into thin films. The resulting films were then analyzed by micro-FTIR with a clean IR-reflective brass plate used as a background.

METHOD: (Covered Under A2LA Accreditation)

A Fourier Transform Infrared Spectrometer was used to collect and process infrared wavelength absorbance / transmission spectra of the specimens. These spectra are created when a molecule converts infrared radiation into molecular vibrations. There are two types of molecular vibrations, stretching and bending. These vibrational movements create bands in a spectrum that occur at specific wavelengths (cm^{-1}). Each wavelength is dependent on a number of things. Most importantly of which are the mass of the atoms present, the force constants of the bonds present, and the geometry of the molecule present.

Infrared spectra can help in identifying the chemical composition and / or bonding present in an unknown molecule. The greatest use of infrared spectroscopy is that the tool provides structural information about the presence of certain functional groups that are present in the sample.

The spectrometer radiates a broad band of infrared light through the specimen. Depending on the bonding present, the light will be absorbed, transmitted, or reflected at various wavelengths. From the spectrum produced, information about the bonding present is obtained from the location of group frequency peaks. Also, most spectra contain additional “fingerprint” peaks that are unique to a particular molecular structure.



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All molecules have a unique spectrum in IR. The “fingerprint” region, below ~ 1500 cm^{-1} , is typically used to identify the molecule or molecules present. This region is different when comparing two molecules even with identical group frequency peaks. The group frequency peaks are typically strong and convey the presence of numerous organic functional groups. These include, but are not limited to, alcohols (-O-H), amines (-N-H), carbonyls (-C=O), etc.

The spectrometer has the ability to analyze samples in two distinctly different ways. The methods are referred to as bench analysis and micro-FTIR analysis.

Bench analysis consists of forming the specimen to be analyzed into a thin film between two zinc selenide (ZnSe) windows. The infrared beam is then passed through the windows and a spectrum is recorded. This method is typically used on homogenous liquid samples.

Micro-FTIR analysis consists of using a microscope attachment on the spectrometer to analyze areas as small as 25 microns in diameter. This technique is particularly useful in analyzing small areas of surface contamination (given the proper sample geometry) or microscopic materials. Specimens can be analyzed in reflective or transmission mode with this method. In reflective mode, the infrared beam is passed through the sample and then reflected off of a reflective IR plate and passed back through the sample where it is detected and analyzed. In transmission mode, the infrared beam is passed through the sample where it is detected and analyzed. Both modes allow for in situ analysis, limited by the specimen’s geometry. In addition, this type of analysis is typically performed on specimen residues or when a precise location analysis is required.

A specimen’s spectrum can then be compared with spectra of standard materials from an IR reference library or compared with customer-supplied references. Upon analysis of some specimens, a spectrum is obtained that is nothing more than an erratic signal with no distinguishable features. A spectrum of this type is referred to as a noise spectrum. Noise is created in a specimen’s spectrum when the absorbance by the specimen of a particular frequency is extremely low. With extremely low absorbance values, any minor increase or decrease in the absorbance of a frequency can create what appears to be a peak. Primary absorbance peaks lower than approximately 0.4 are generally considered indistinguishable in comparison to any noise present. Additionally, low absorbance values of primary absorbance peaks cause comparisons and identifications to be extremely difficult.

The photograph below shows the FTIR unit.





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Mattson Fourier Transform Infrared Spectrometer with Microscope Attachment

RESULTS:

Overall, a comparison of the FTIR spectra obtained from each submitted sample, suggests there is no substantial difference in chemical structure from sample to sample.

The only real variations observed were slight intensity and peak location shifts for the peaks located just above 1500 cm^{-1} . Based on the fact that the samples are known to be 2-part epoxy systems, this result is not unexpected given that slight variations in the mixture ratio would cause slight variations in peak size and location. Additionally, given the unknown nature of the “unknown” samples’ thermal and chemical exposure histories, these factors could also influence the results observed.

Based on the results of the FTIR analysis alone, no apparent significant differences were detected between the “unknown” and “known” samples. All appear to be, chemically speaking, the same.

All of the sample spectra obtained can be found attached at the end of the report.





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DIFFERENTIAL SCANNING CALORIMETRY (DSC) ANALYSIS

TEST SPECIMENS:

Five (5) “unknown” and four (4) “known” epoxy samples, identified as:

“Unknown” Samples

- EB081 M1 2/3/4 NW
- EB081 M2 2/3/4 SE
- WB082 S 2 S
- WB083 M 9/10/11 SW
- EB080 N 4/5/6 NE

“Known Samples”

- Std 50/50
- Std 70/30
- Fast 50/50
- Fast 70/30

SPECIMENS PREPARATION:

For each submitted epoxy sample, a small specimen was removed and was mounted in a standard aluminum specimen holder with a lid. The holder and lid were then compressed around the specimen in order to seal the container.

TEST PARAMETERS:

Start Temperature	=	- 65°C
Final Temperature	=	+ 100°C
Purge Gas	=	Nitrogen
Scanning Rate	=	20°C/min

METHOD:

Each specimen was tested in order to assign the Glass Transition Temperature (T_g) and to determine the Cure Factor (ΔT_g). The T_g is generally considered the minimum temperature for curing to proceed. The ΔT_g is an indication of cure and degradation.

This method subjects the specimen to a precisely controlled temperature environment while continuously monitoring the heat input and output. Reactions within the specimen cause changes in the heat input and output.

Each specimen was individually placed in a Differential Scanning Calorimeter and curves of Temperature vs. Heat Flow were generated. Upon completion of the curve, the sample was held at the final temperature for 15 minutes and then cooled for approximately 5 minutes to - 65°C with the temperature cycle then repeated.

The glass transition is an endothermic process that produces an increase in the slope of the Temperature vs. Heat Flow curve. The glass transition temperature is the temperature



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chosen to represent the range over which the glass transition occurs. This temperature (T_g) can either be the point that corresponds to the maximum value of the slope during this change or the approximate midpoint of the temperature range over which the glass transition occurs.

These temperatures were determined for both runs with the difference between the T_g values of both runs defined as the ΔT_g .

RESULTS:

From the analysis, several of the submitted samples provided a less than “ideal” DSC thermogram. This result is a function of the sample itself. Given these somewhat erratic scans, the most logical point of T_g was chosen for each sample scan. Table 1 summarizes the results of the DSC Analysis.

(Note: As was the case for the FTIR results given above, the unknown nature of the “unknown” samples’ thermal and chemical exposure histories along with small variations due to the epoxy mixture, could affect the results obtained for the DSC analysis when trying to compare results to those obtained for the “known” samples.)

Table 1: DSC Analysis Results

	Sample ID	1 st Scan (°C)	2 nd Scan (°C)	ΔT_g (°C)
“Unknown”	EB081 M1 2/3/4 NW	61.3	60.3	-1.0
	EB081 M2 2/3/4 SE	69.4	92.7	23.3
	WB082 S 2 S	41.0	57.1	16.1
	WB083 M 9/10/11 SW	70.3	68.0	-2.3
	EB080 N 4/5/6 NE	5.0	11.4	6.4
“Known”	Std 50/50	40.4	41.0	0.6
	Fast 50/50	61.3	49.0	- 12.3
	Std 70/30	-10.3	-9.0	1.3
	Fast 70/30	-9.3	-1.7	7.6

Based on the results found in the table above, the following observations were made:

- The 1st Scan T_g values for the “Std 50/50” and “Fast 50/50” samples shows a distinct “mechanical” difference
- The sub-ambient T_g values found for the “Std 70/30” and “Fast 70/30” samples was consistent with the “gel-like” consistency of the samples at laboratory ambient conditions.



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- The “WB082 S 2 S” sample appears to be most similar to the “Std 50/50” sample based on the Tg value obtained from the 1st scan
- The “EB081 M1 2/3/4 NW”, “EB081 M2 2/3/4 SE”, and “WB083 M 9/10/11 SW” samples appear to be most similar to the “Fast 50/50” sample based on the Tg value obtained from the 1st scan
- None of the samples appear to be similar to the “Std 70/30” or “Fast 70/30” samples
- The 1st scan Tg value found for the “EB080 N 4/5/6 NE” sample suggests that it may be a component mixture somewhere between “50/50” and “70/30”
- The various ΔT_g values found suggest varying levels of cure or possibly effects due to thermal or chemical exposure

All of the sample DSC scans obtained can be found attached at the end of the report.





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THERMOGRAVIMETRIC ANALYZER (TGA) ANALYSIS

TEST SPECIMENS:

Three (3) “unknown” and two (2) “known” epoxy samples, identified as:

“Unknown” Samples

- EB081 M1 2/3/4 NW
- EB081 M2 2/3/4 SE
- WB082 S 2 S

“Known Samples”

- Std 50/50
- Fast 50/50

TEST PARAMETERS:

Temperature Range	=	30°C to 900°C
Atmosphere	=	Nitrogen
Scanning Rate	=	10.0°C/min

METHOD:

Each submitted sample was analyzed in the same manner. A 10 mg (nominal) specimen of the material was placed in a sample pan attached to a microbalance. Using the Thermogravimetric Analyzer, the specimen was heated according to the thermal profile listed above, in a nitrogen atmosphere with 20 to 30 mL/min gas flow.

The test method subjects the sample to a precisely controlled temperature environment while continuously monitoring its change in weight. The weight and the derivative weight of each sample were recorded on each specific scan.

RESULTS:

For each sample tested, four (4) decomposition “points” were found along with the “final” point in which the only material left was filler / carbon black.

Tables 2 and 3, below, summarize the results of the TGA Analysis. Each table gives the temperature and “% sample remaining” values for each “point”, along with the overall filler content value at the end of the test.

(Note: As was the case for the FTIR and DSC results given above, the unknown nature of the “unknown” samples’ thermal and chemical exposure histories along with small variations due to the epoxy mixture, could affect the results obtained for the DSC analysis when trying to compare results to those obtained for the “known” samples.)





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Table 2: TGA Analysis Results – “Unknown” Samples

“Points”	EB081 M1 2/3/4 NW		EB081 M2 2/3/4 SE		WB 082 S 2 S	
	(°C)	(%)	(°C)	(%)	(°C)	(%)
1	92.2	98.8	160.2	97.7	155.1	97.8
2	279.1	88.1	349.0	86.3	345.4	86.2
3	454.1	67.2	517.9	57.3	517.6	57.8
4	675.9	43.4	681.3	42.6	687.1	42.9
Final	752.6	32.1	749.2	31.9	742.4	32.4
Filler Content	31.8 %		31.5 %		32.0 %	

Table 3: TGA Analysis Results – “Known” Samples

“Points”	Std 50/50		Fast 50/50	
	(°C)	(%)	(°C)	(%)
1	116.4	99.6	108.4	99.7
2	352.1	88.0	348.5	84.9
3	506.0	61.5	534.3	62.3
4	681.4	53.2	695.2	48.5
Final	775.8	34.0	782.9	31.0
Filler Content	34.0 %		30.7 %	

Based on the results found in the table above, the following observations were made:

- The “EB081 M2 2/3/4 SE” and “WB 082 S 2 S” samples showed almost identical thermal decomposition characteristics over the course of the test, for both temperature, “% sample remaining”, and filler content
- The “Std 50/50” and “Fast 50/50” samples also showed very similar thermal decomposition characteristics over the course of the test
- Overall, the thermal decomposition characteristics of all of the samples, except “EB081 M1 2/3/4 NW”, were similar with the two (2) “unknown” samples decomposing slightly faster than the two (2) “known” samples
- The characteristics of the “EB081 M1 2/3/4 NW” sample failed to match any of the other “unknown” or “known” samples analyzed, specifically a much lower “Point #3” temperature was found and a significantly different “% sample remaining” profile from “Point #2” to “Point #4” was measured

All of the sample TGA scans obtained can be found attached at the end of the report.



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1251.03

ISO 9001:2000



ISO/IEC 17025



Cert No. C2005-01504

Trace Laboratories-East certifies that the test equipment used complies with the calibration test purposes of ISO 10012-1, ANSI/NCSL Z540-1-1994, and MIL-STD-45662A and that the data contained in this report is accurate within the tolerance limitation of this equipment.

All test procedures detailed within this report are complete. The results in this report relate only to those items tested. If any additional information or clarification of this report is required, please contact us. This test report shall not be reproduced except in full, without the written approval of Trace Laboratories-East.

Thank you for selecting Trace Laboratories-East for your testing purposes.

PERFORMED BY:

[Redacted Signature]

Keith M Sellers
Senior Scientist

[Redacted Signature]

Diane M. Andercyk
Senior Engineer

[Redacted Signature]

Daniel D. Phillips
Engineer

Attachments: eleven (11) FTIR scans
eighteen (18) DSC scans
five (5) TGA scans

SAMPLE DISPOSITION: Samples returned to customer





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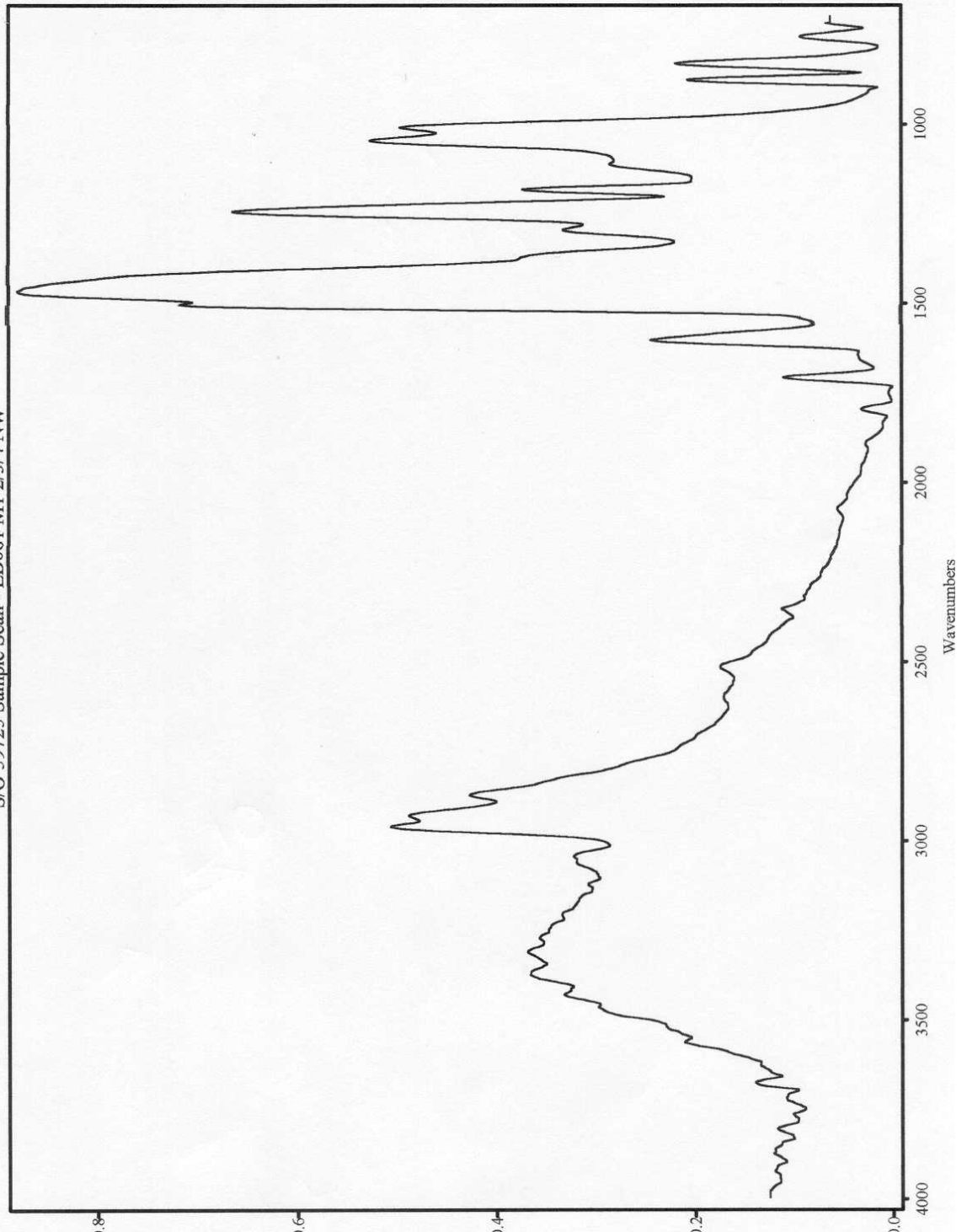


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S/O 59725 Sample Scan - EB081 M1 2/3/4 NW





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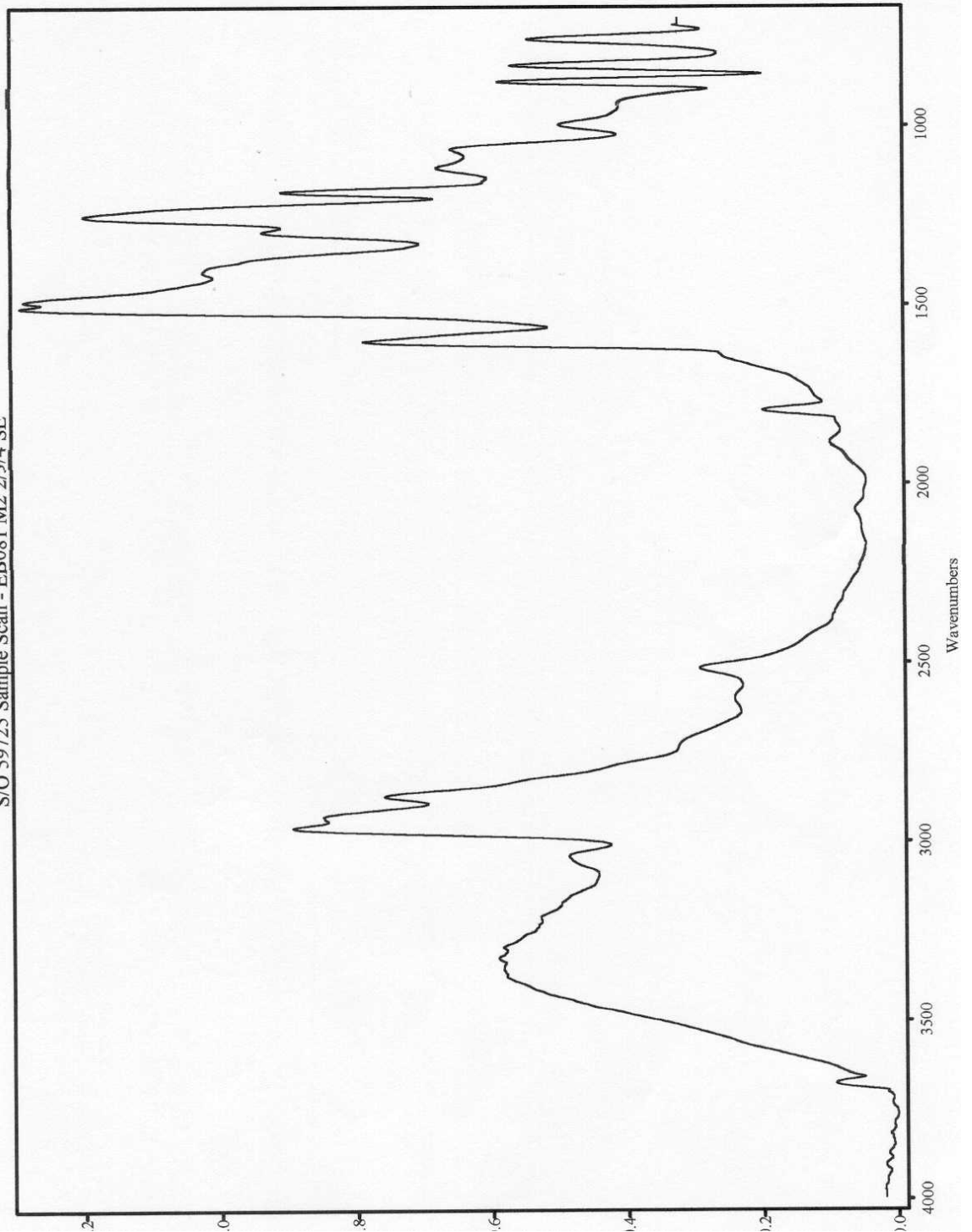


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S/O 59725 Sample Scan - EB081 M2 2/3/4 SE





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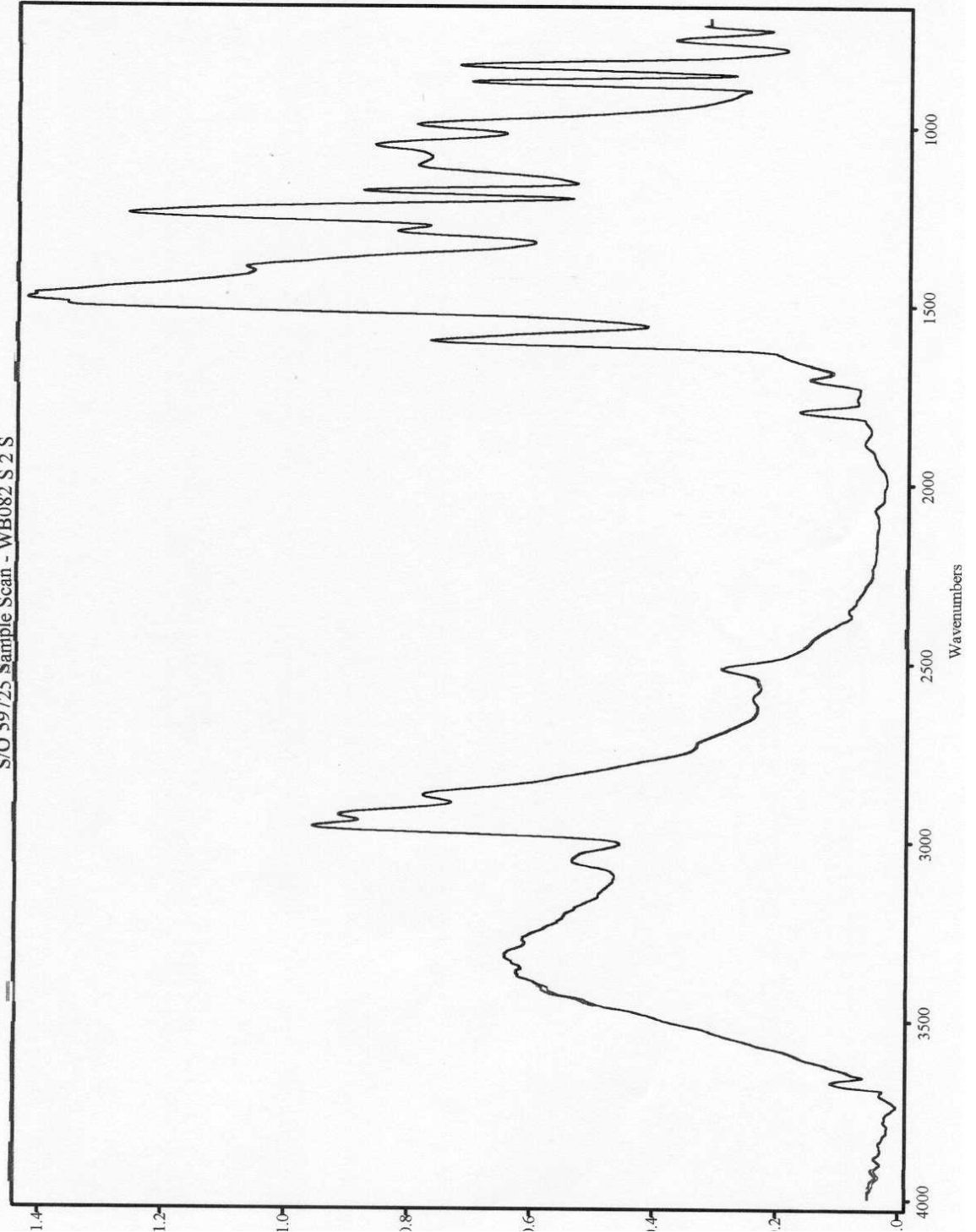
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PERRY JOHNSON
REGISTRARS, INC.

Cert No. C2005-01504

S/O 59725 Sample Scan - WB082 S 2 S





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ISO 9001:2000

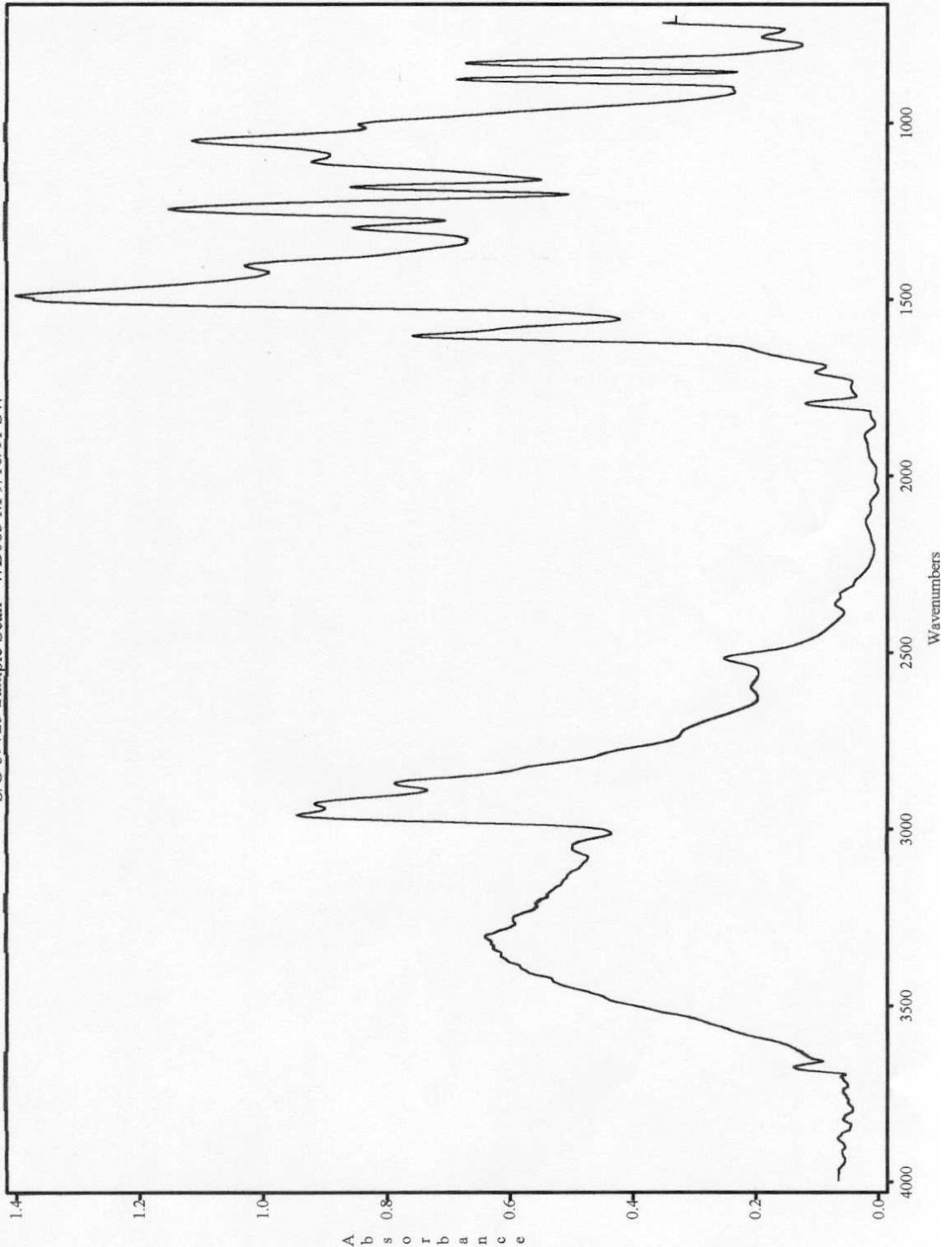


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S/O 59725 Sample Scan - WB083 M 9/10/11 SW





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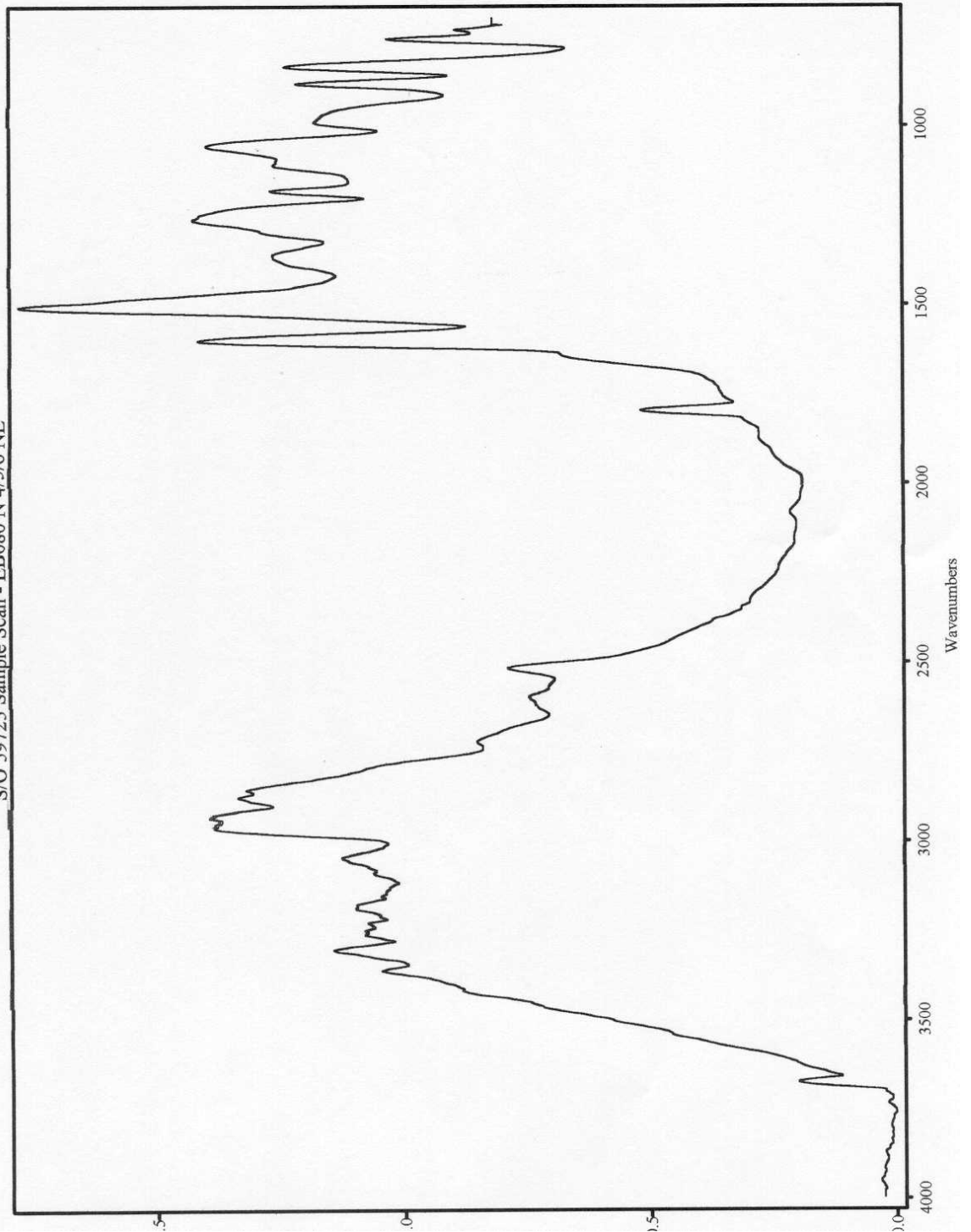


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S/O 59725 Sample Scan - EB080 N 4/5/6 NE





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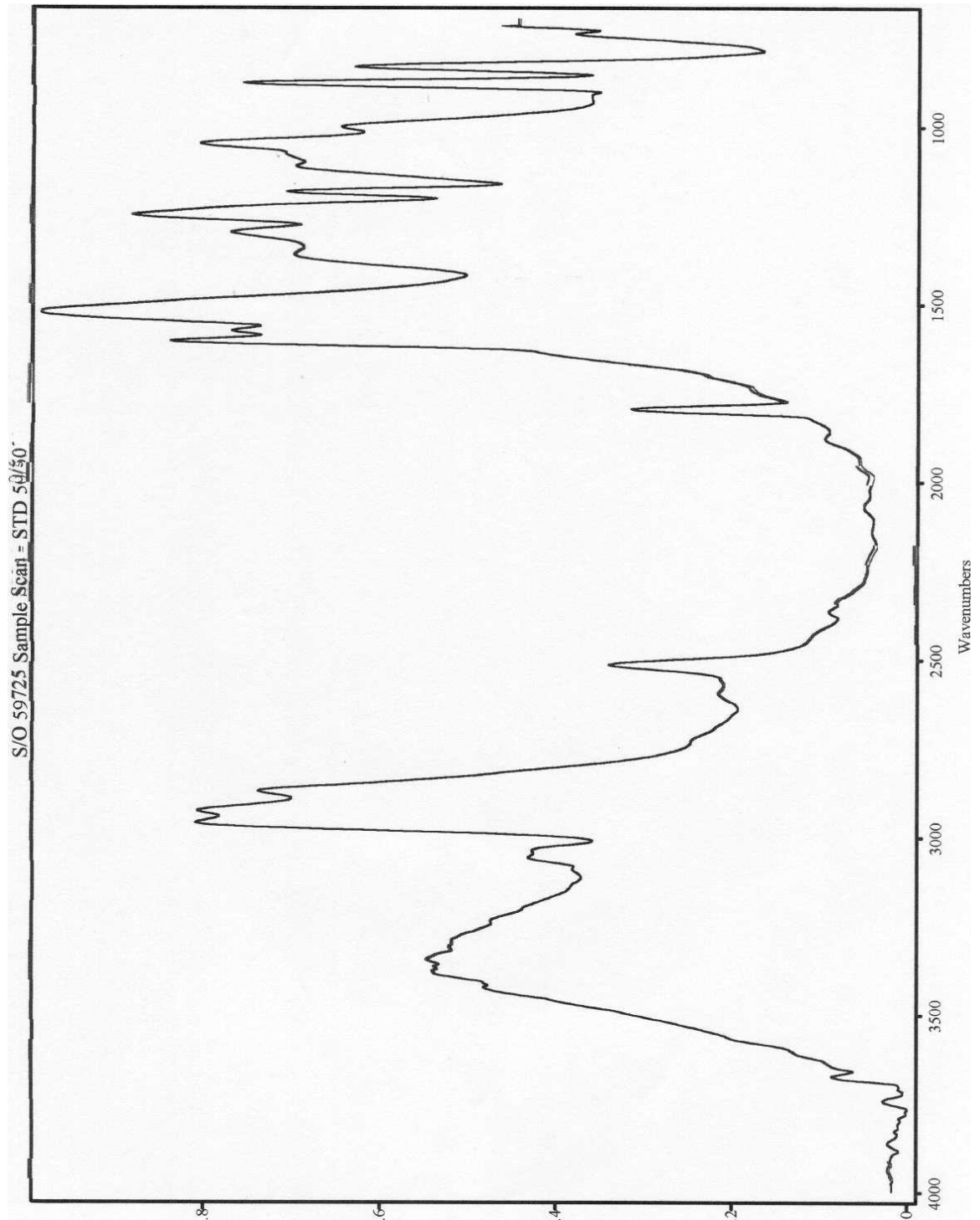
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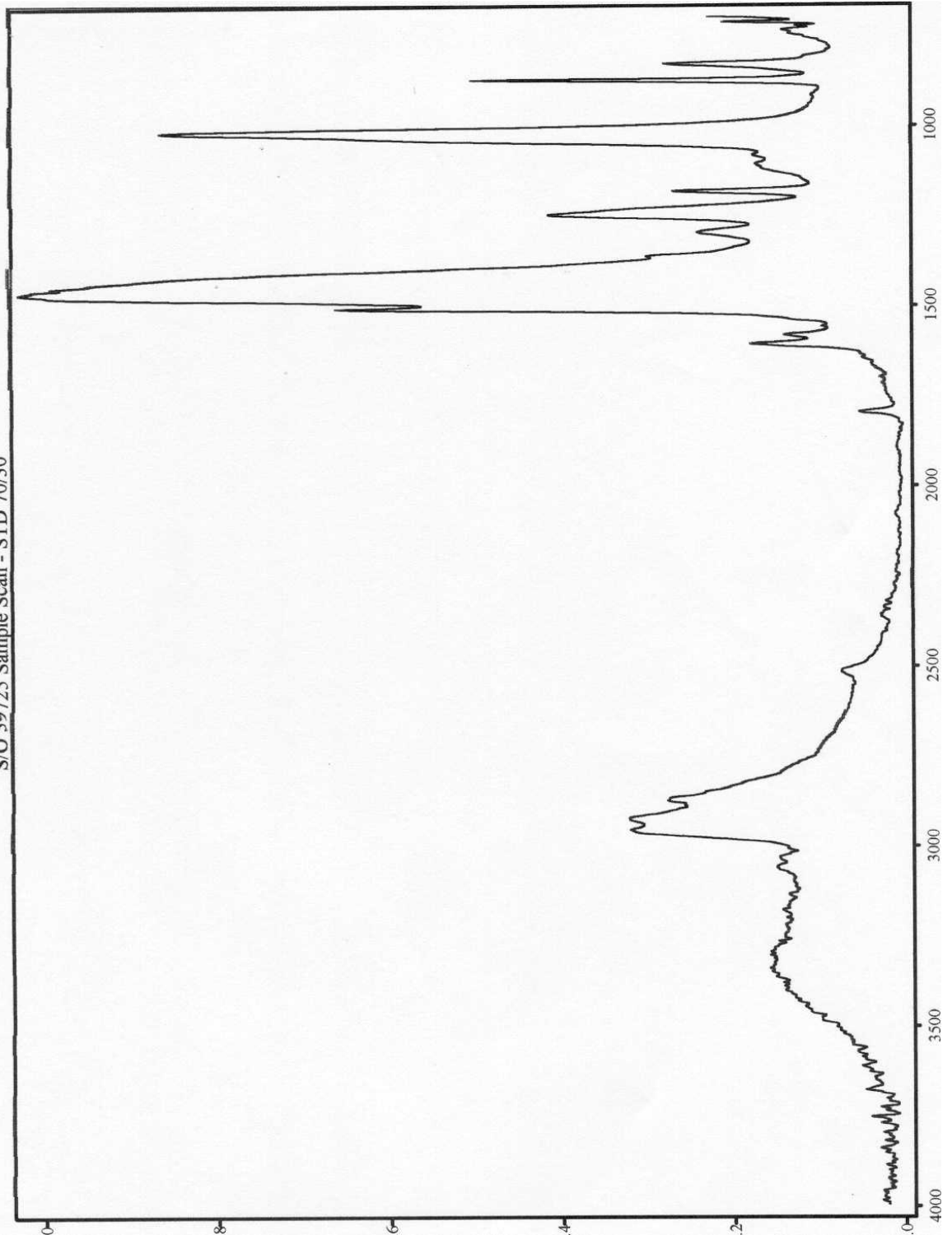


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S/O 59725 Sample Scan - STD 70/30





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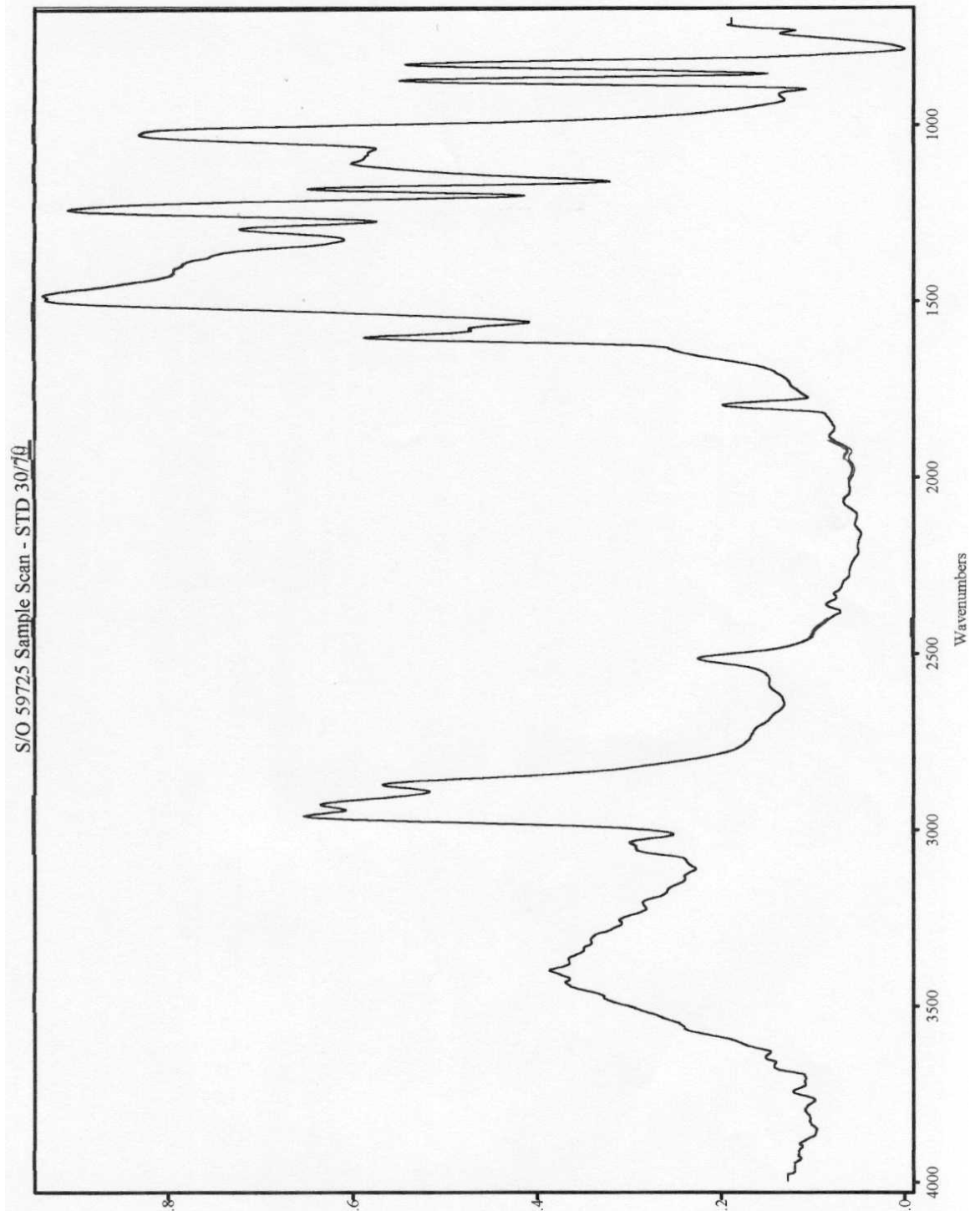
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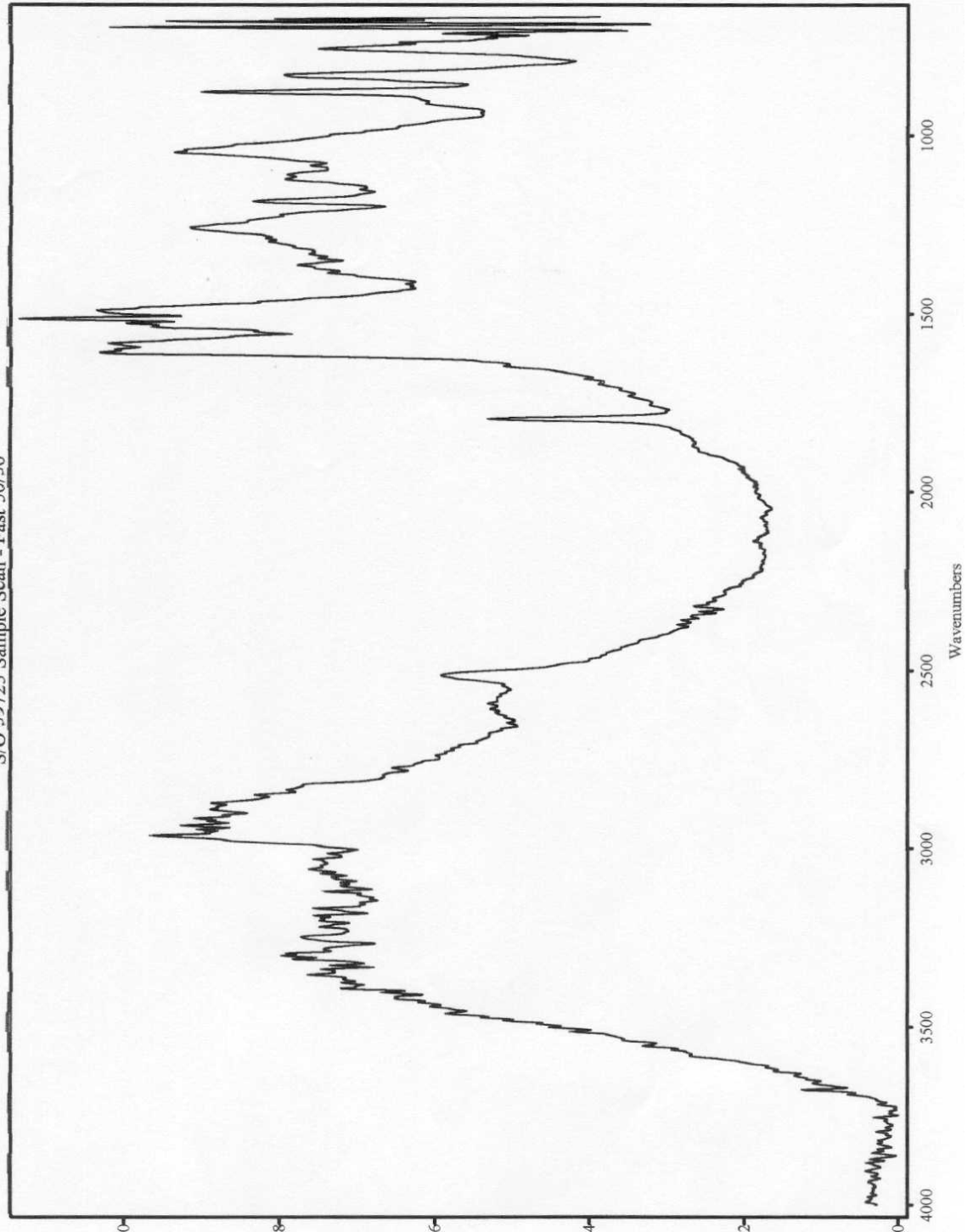


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S/O 59725 Sample Scan - Fast 50/50





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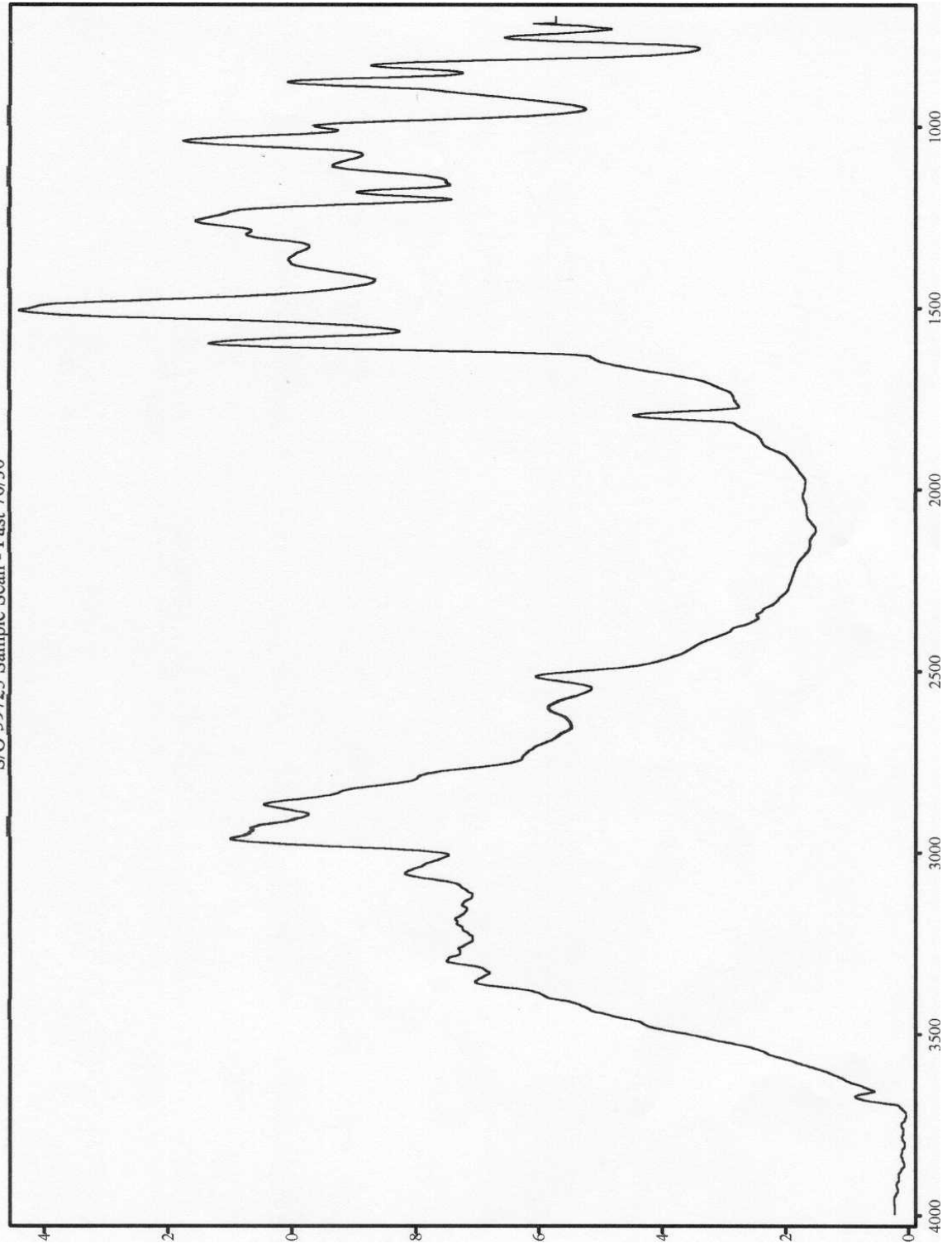


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S/O 59725 Sample Scan - Fast 70/30





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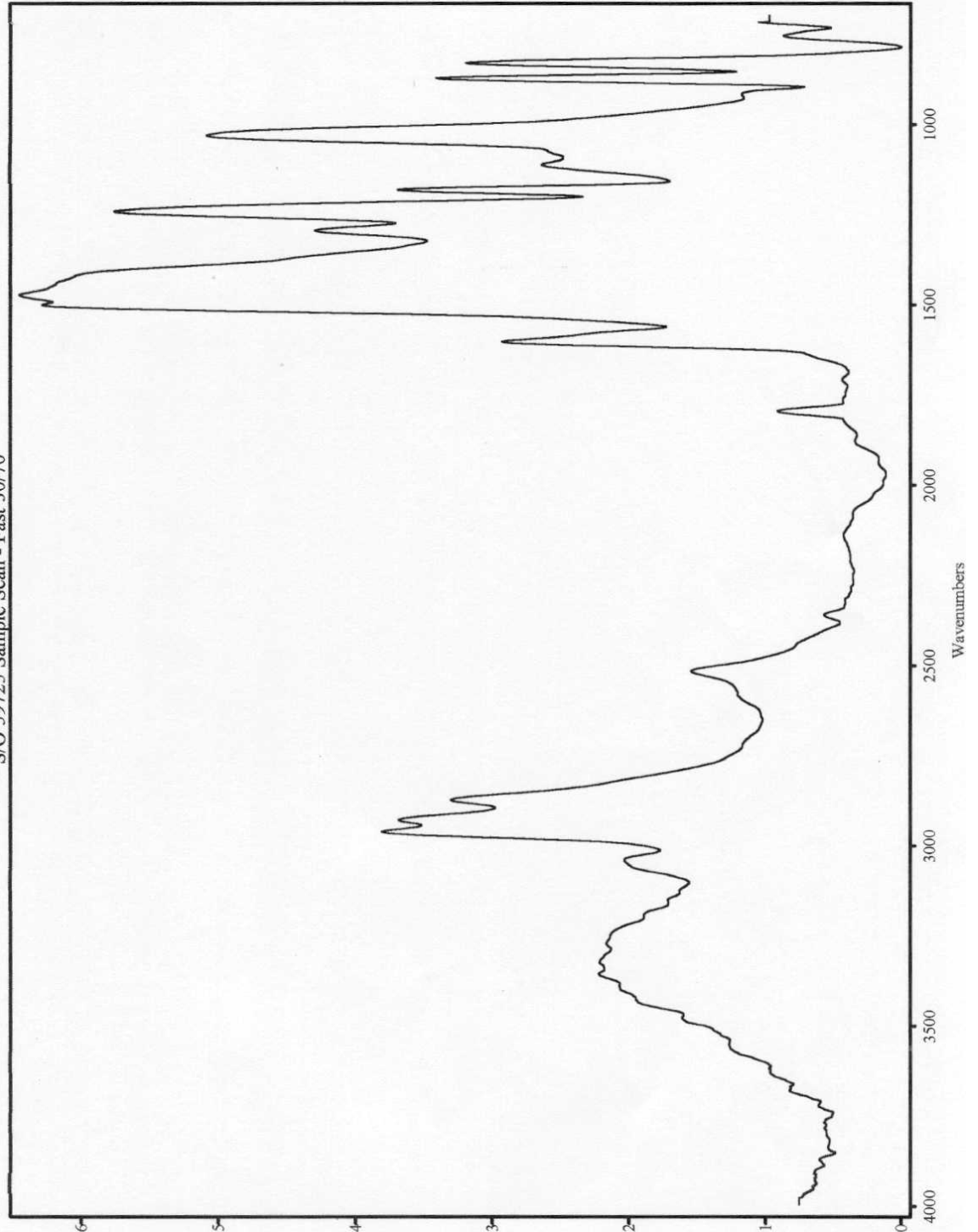


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S/O 59725 Sample Scan - Fast 30/70





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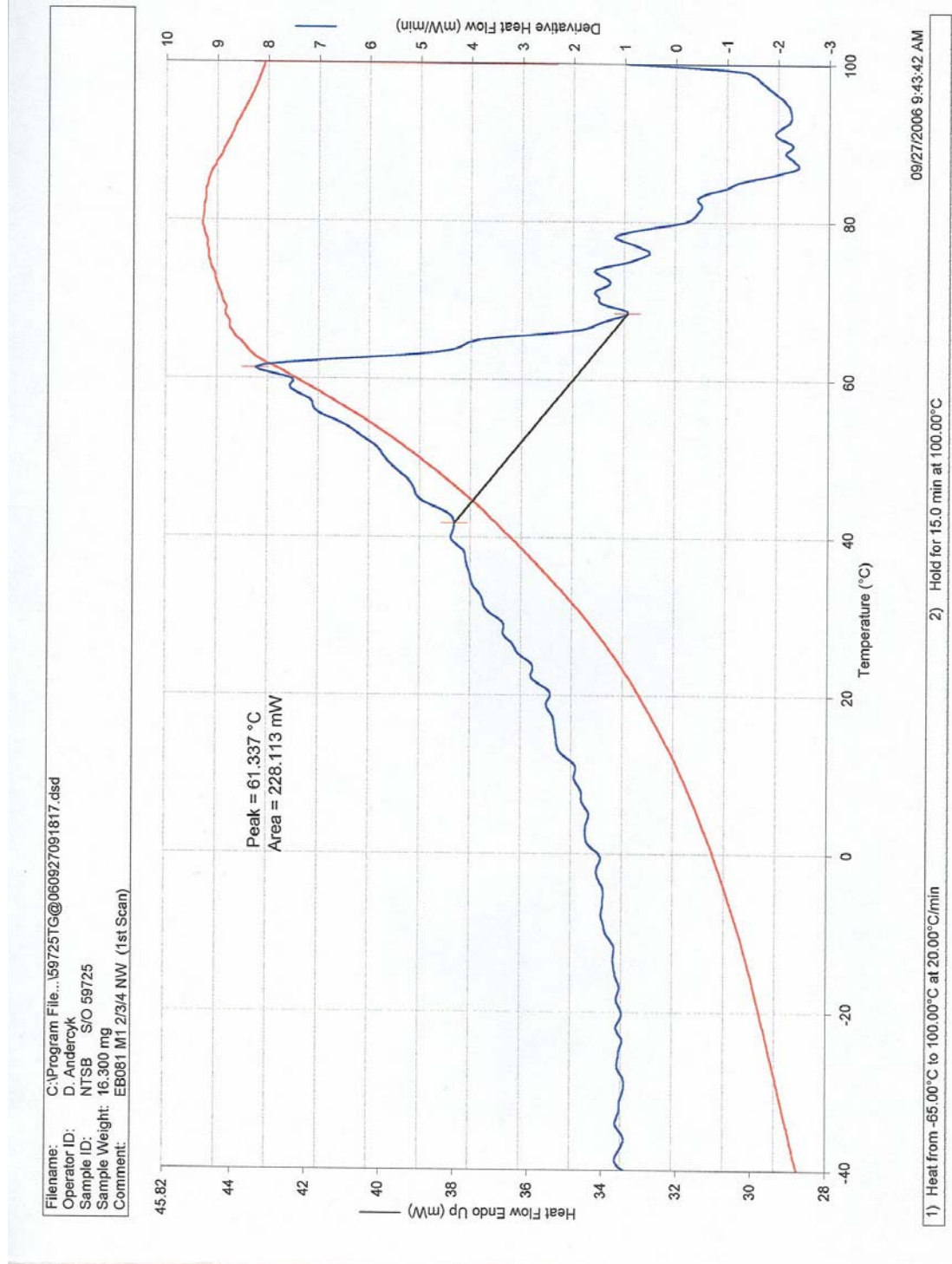
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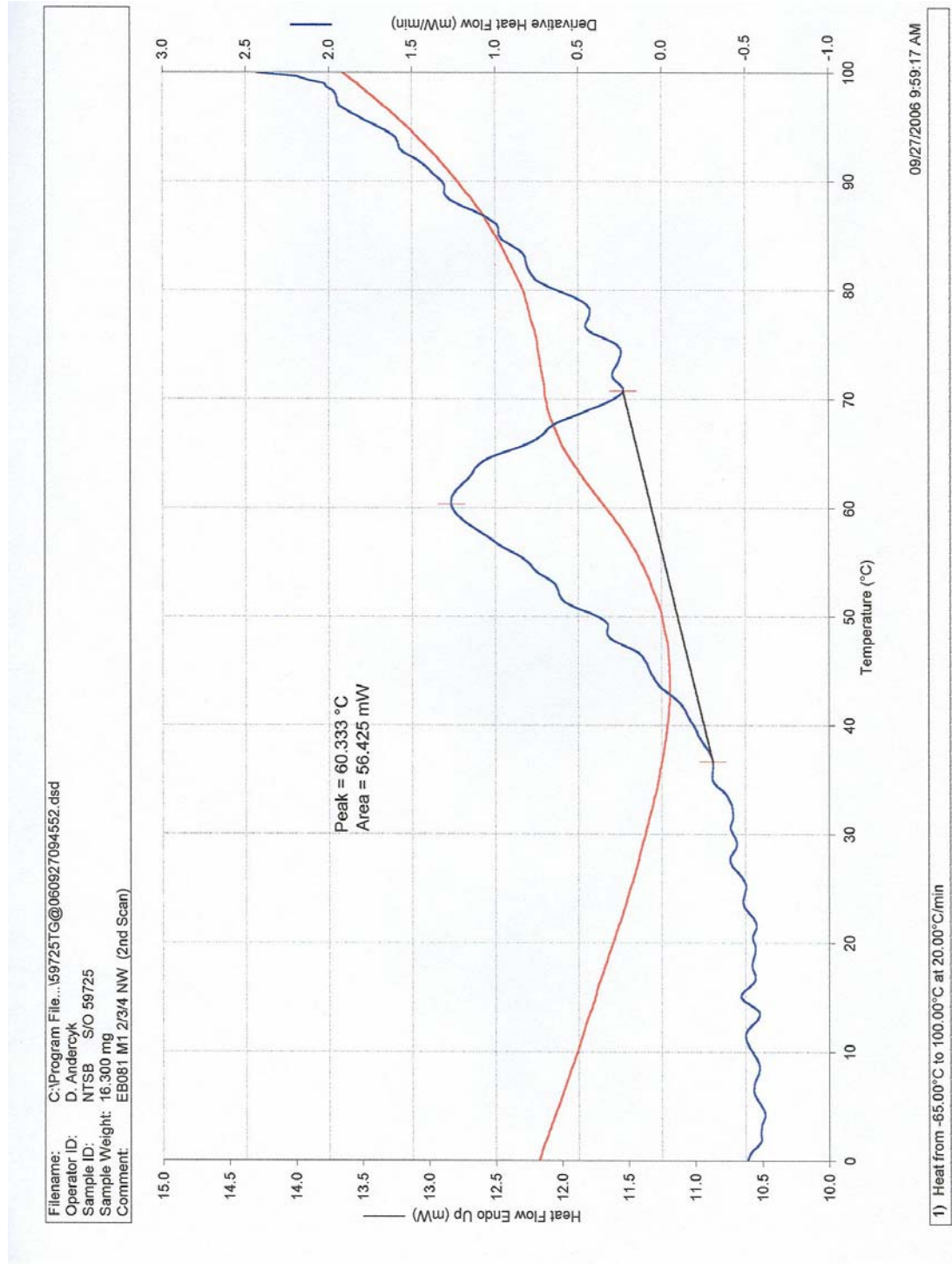
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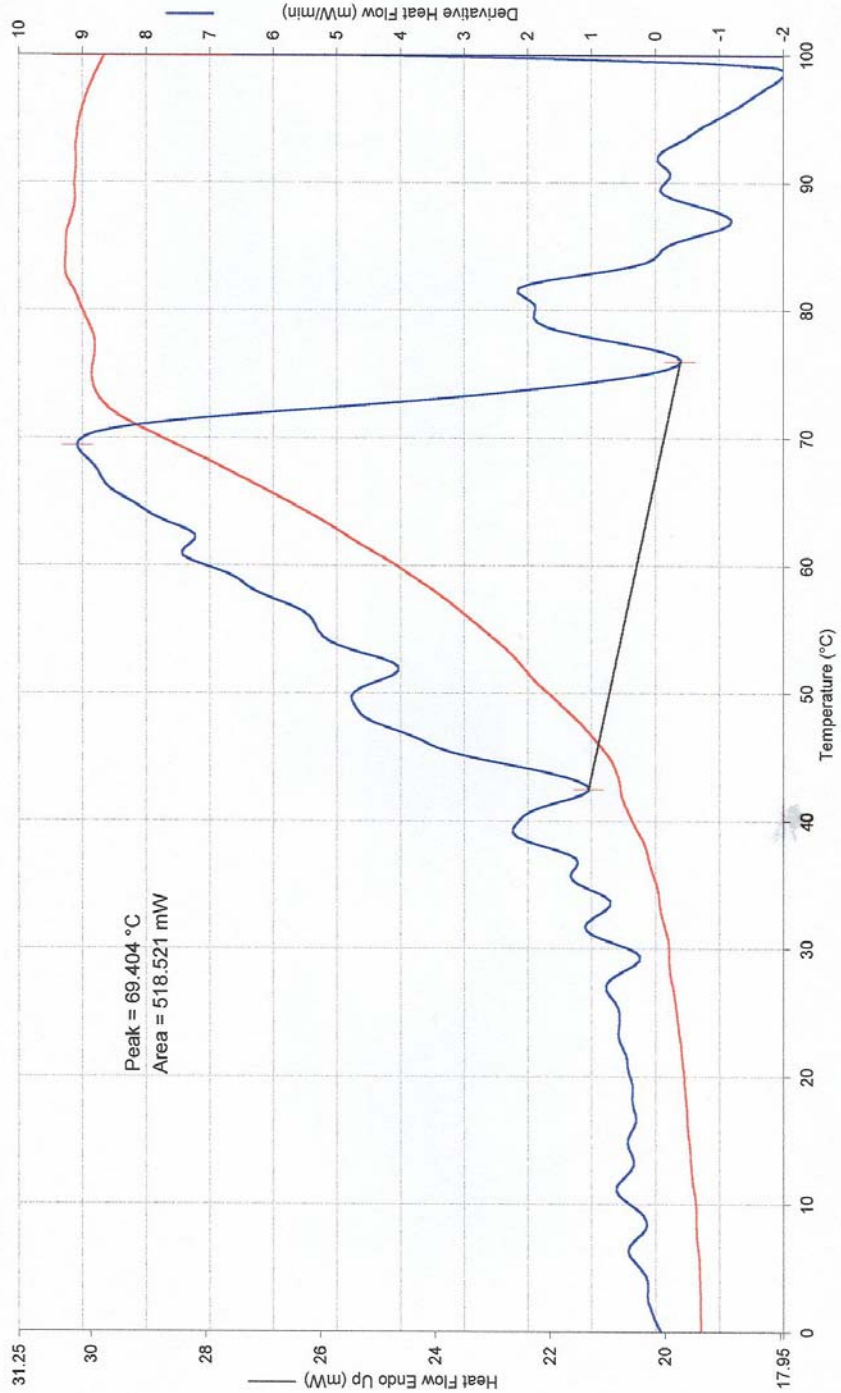


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...59725TG@060927095737.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 18.000 mg
Comment: EB081 M2 2/3/4 SE (1st Scan)



09/27/2006 10:42:46 AM

2) Hold for 15.0 min at 100.00 °C

1) Heat from -65.00 °C to 100.00 °C at 20.00 °C/min





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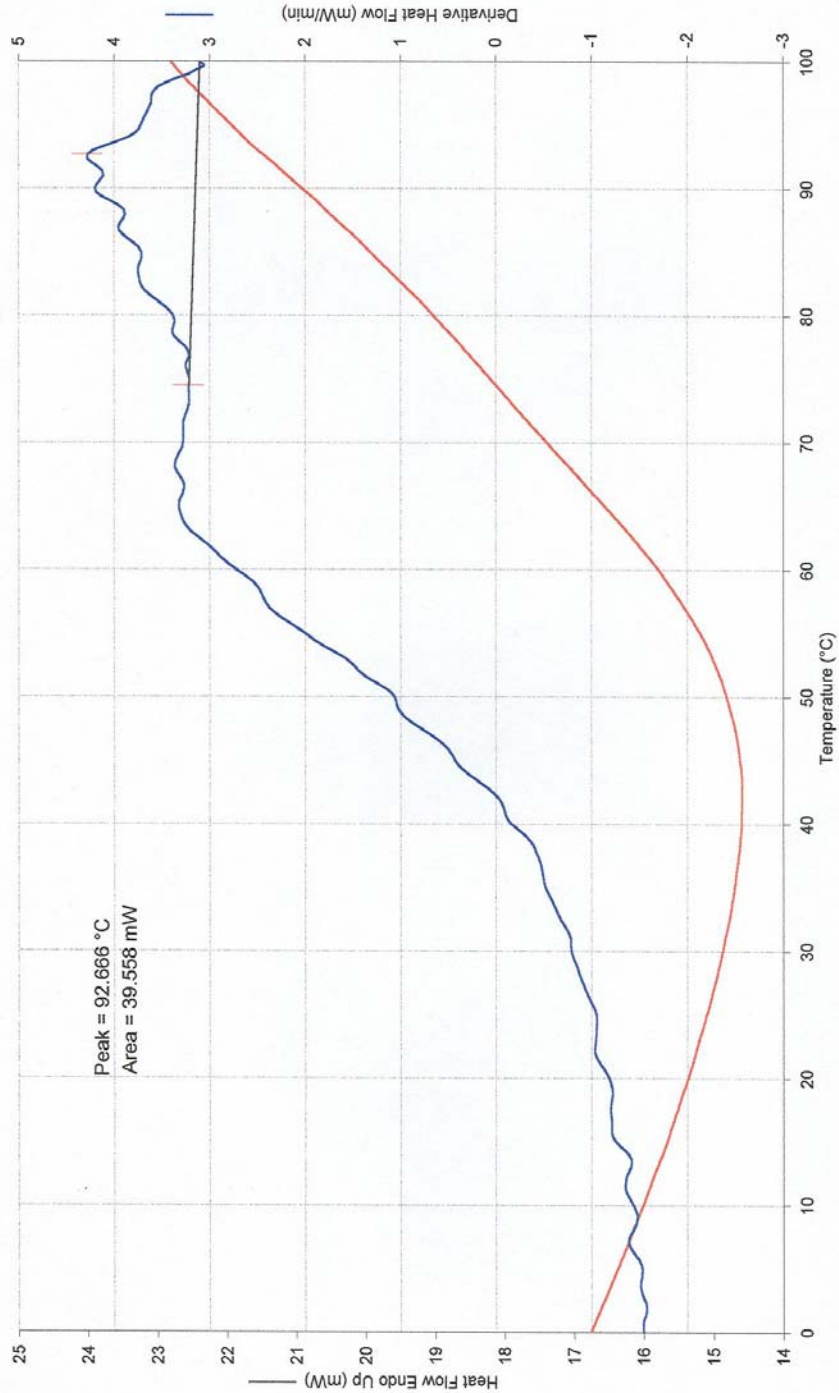


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...59725TG@060927102512.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 18.000 mg
Comment: EB081 M2 2/3/4 SE (2nd Scan)



09/27/2006 10:42:38 AM

[1] Heat from -65.00°C to 100.00°C at 20.00°C/min





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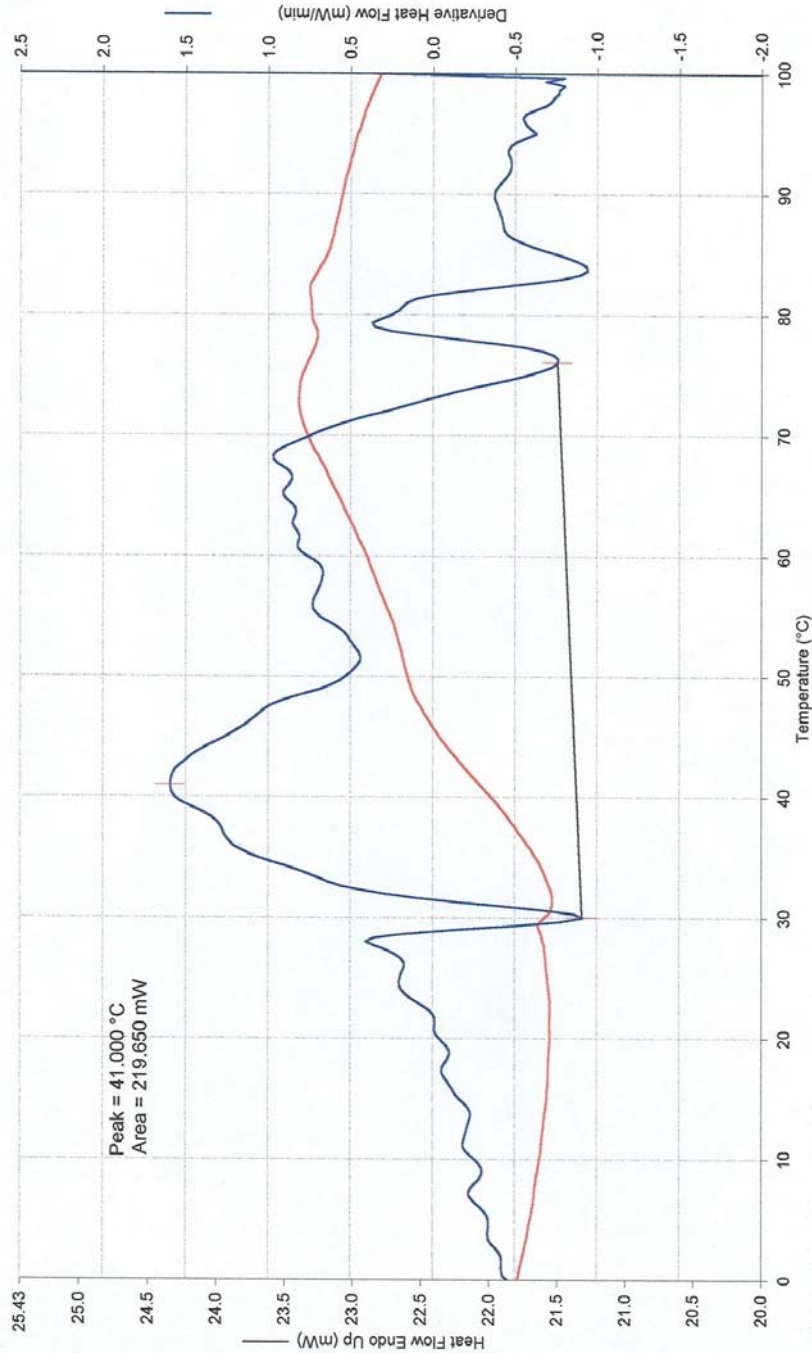


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...\59725TG\060926125420.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 19.300 mg
Comment: WB082 S 2 S (1st Scan)



09/26/2006 1:34:16 PM

2) Hold for 15.0 min at 100.00°C

1) Heat from -85.00°C to 100.00°C at 20.00°C/min





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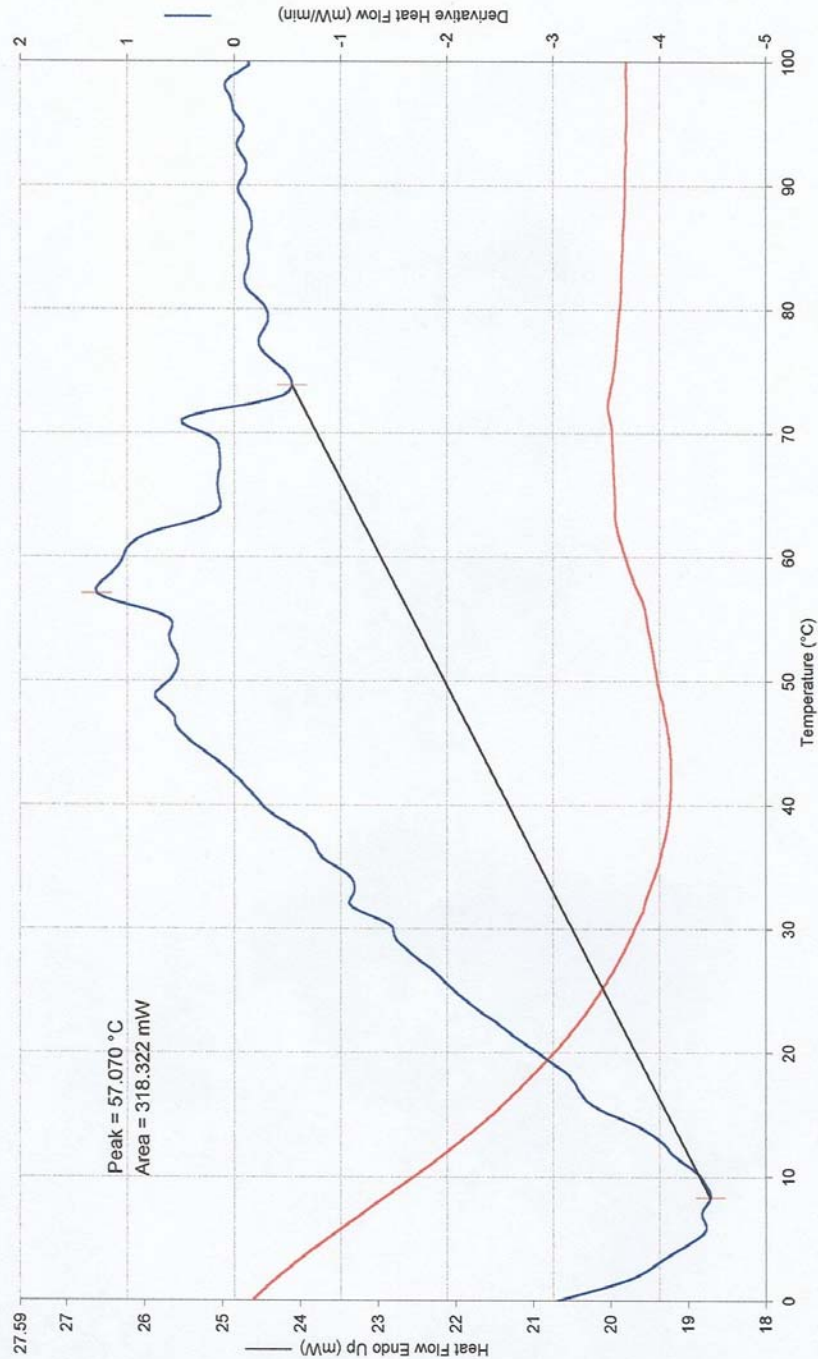


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...59725TG\060926132010.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 19.300 mg
Comment: WB082 S 2 S (2nd Scan)



09/26/2006 1:33:39 PM

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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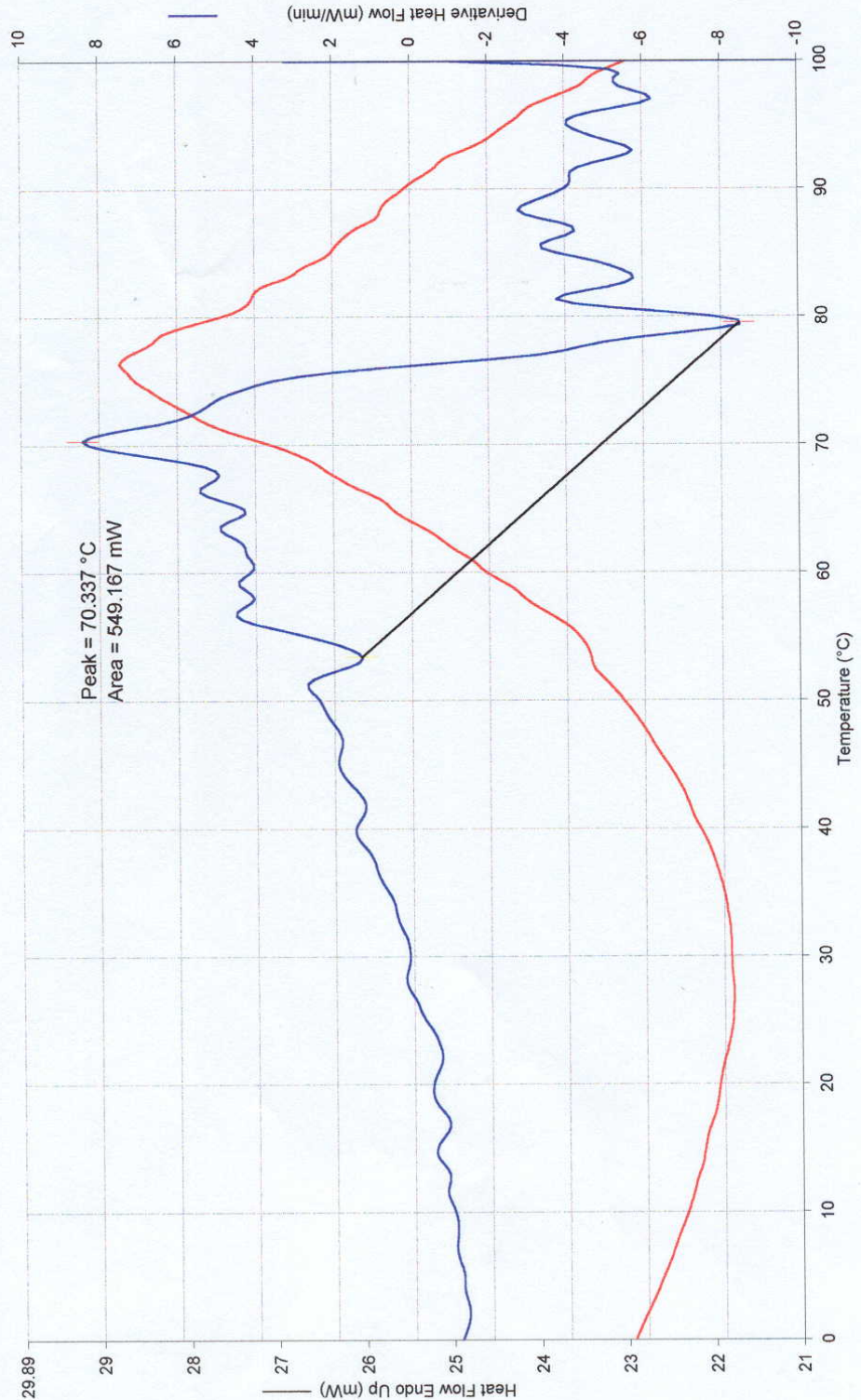


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Cert No. C2005-01504

Filename: C:\Program File...\59725TG@060927075628.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 19.500 mg
Comment: WB083 M 9/10/11 SW (1st Scan)



09/27/2006 8:27:33 AM

2) Hold for 15.0 min at 100.00°C

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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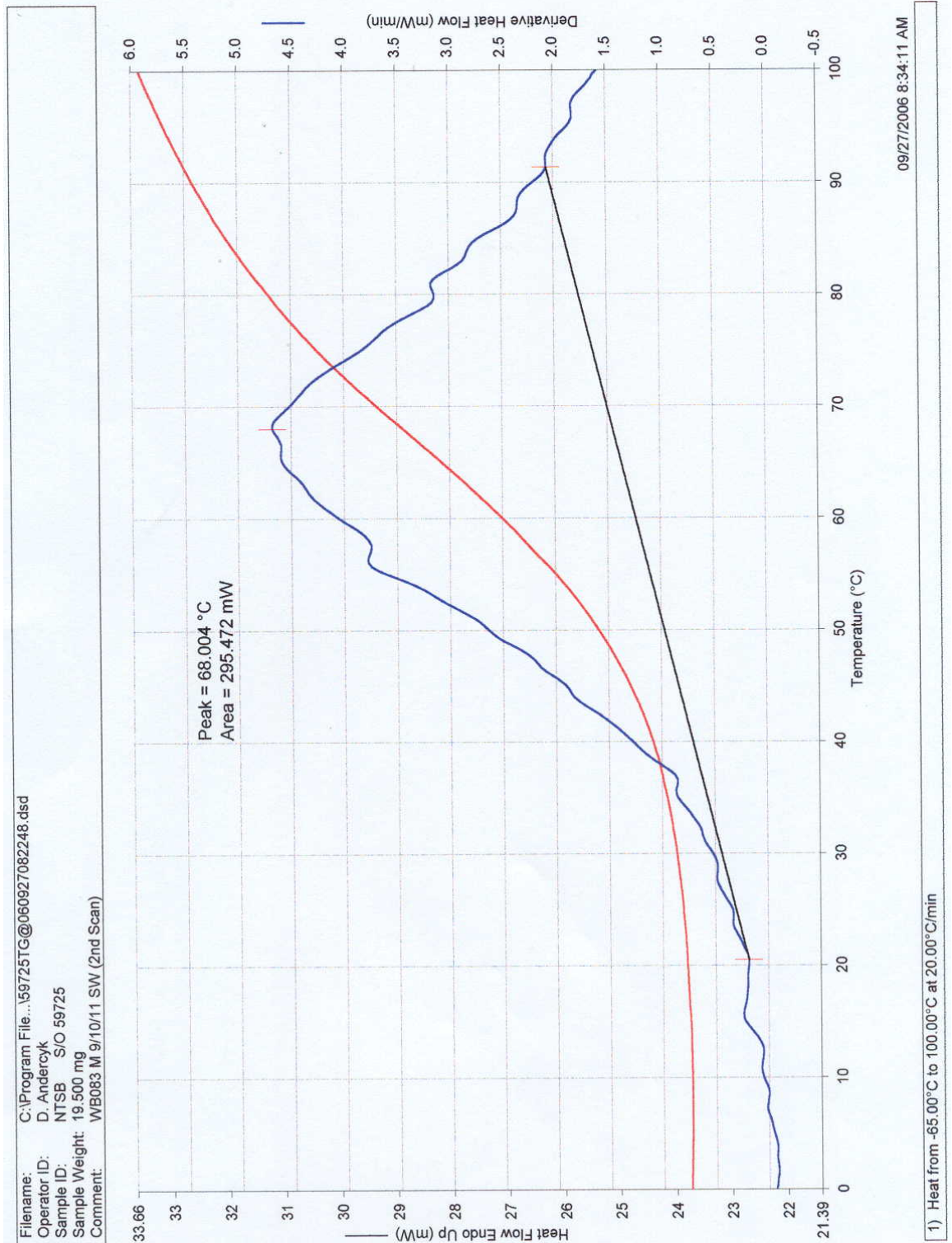
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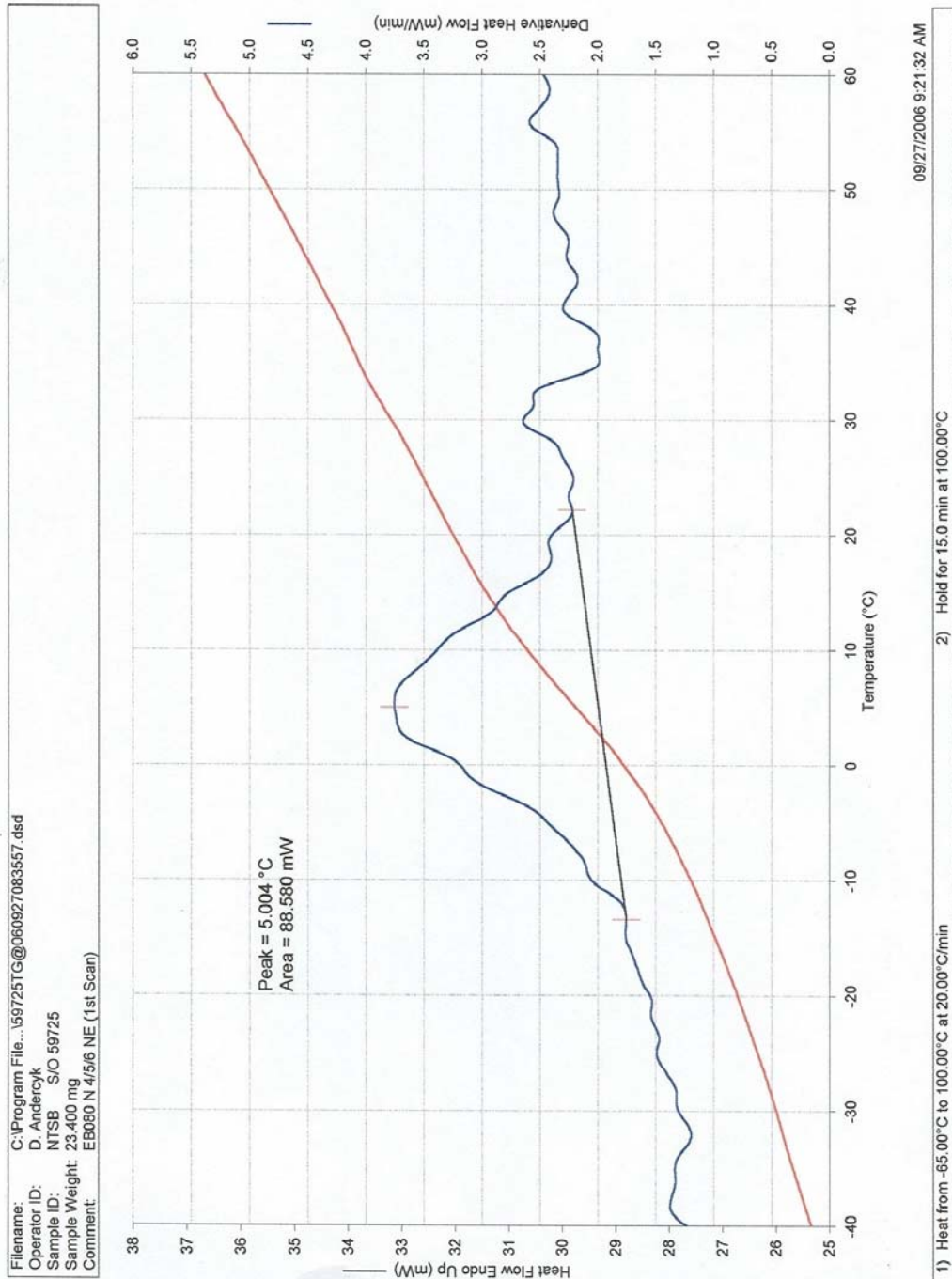
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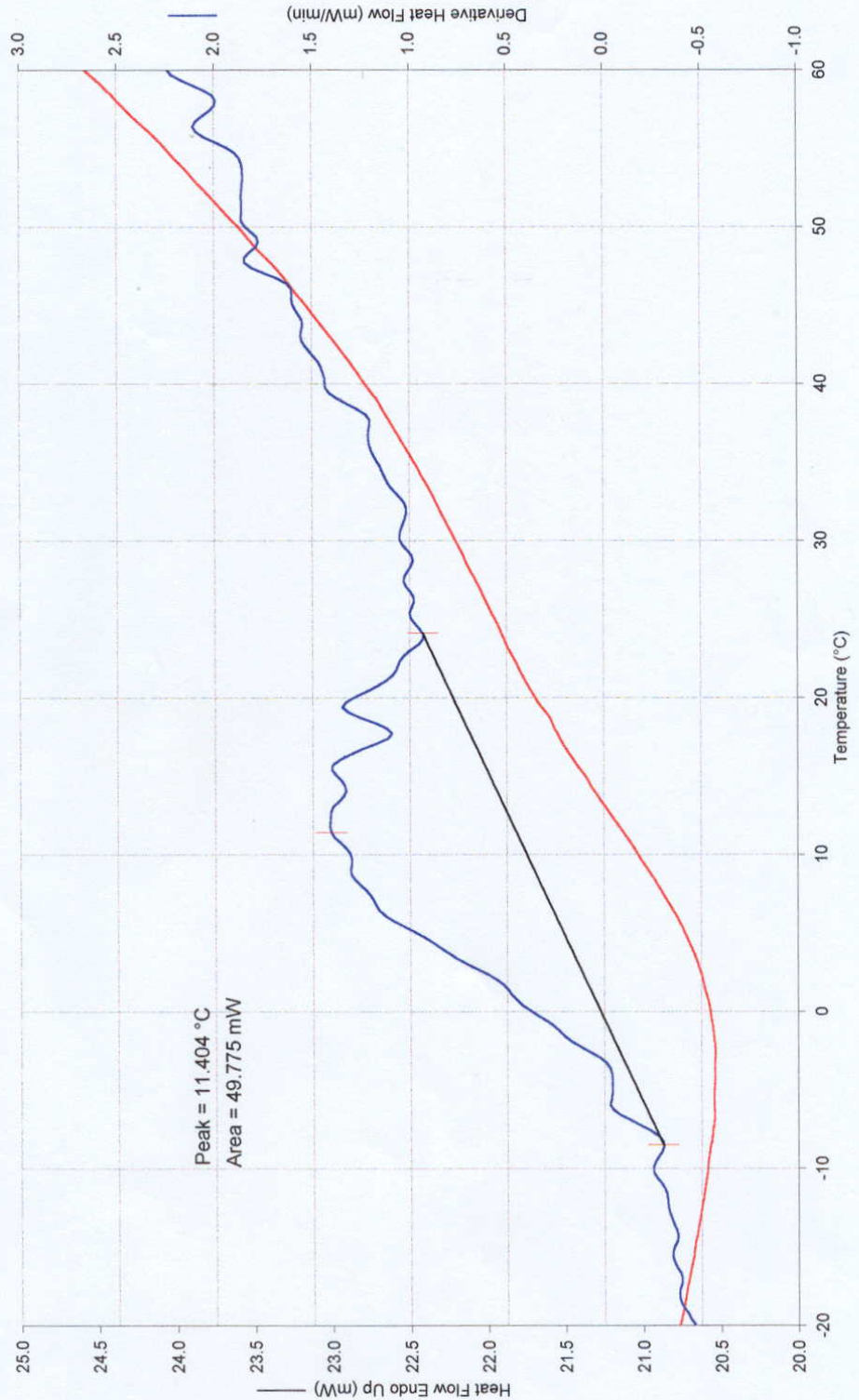


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...\59725TG@060927090450.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 23.400 mg
Comment: EB080 N 4/5/6 NE (2nd Scan)



09/27/2006 9:21:43 AM

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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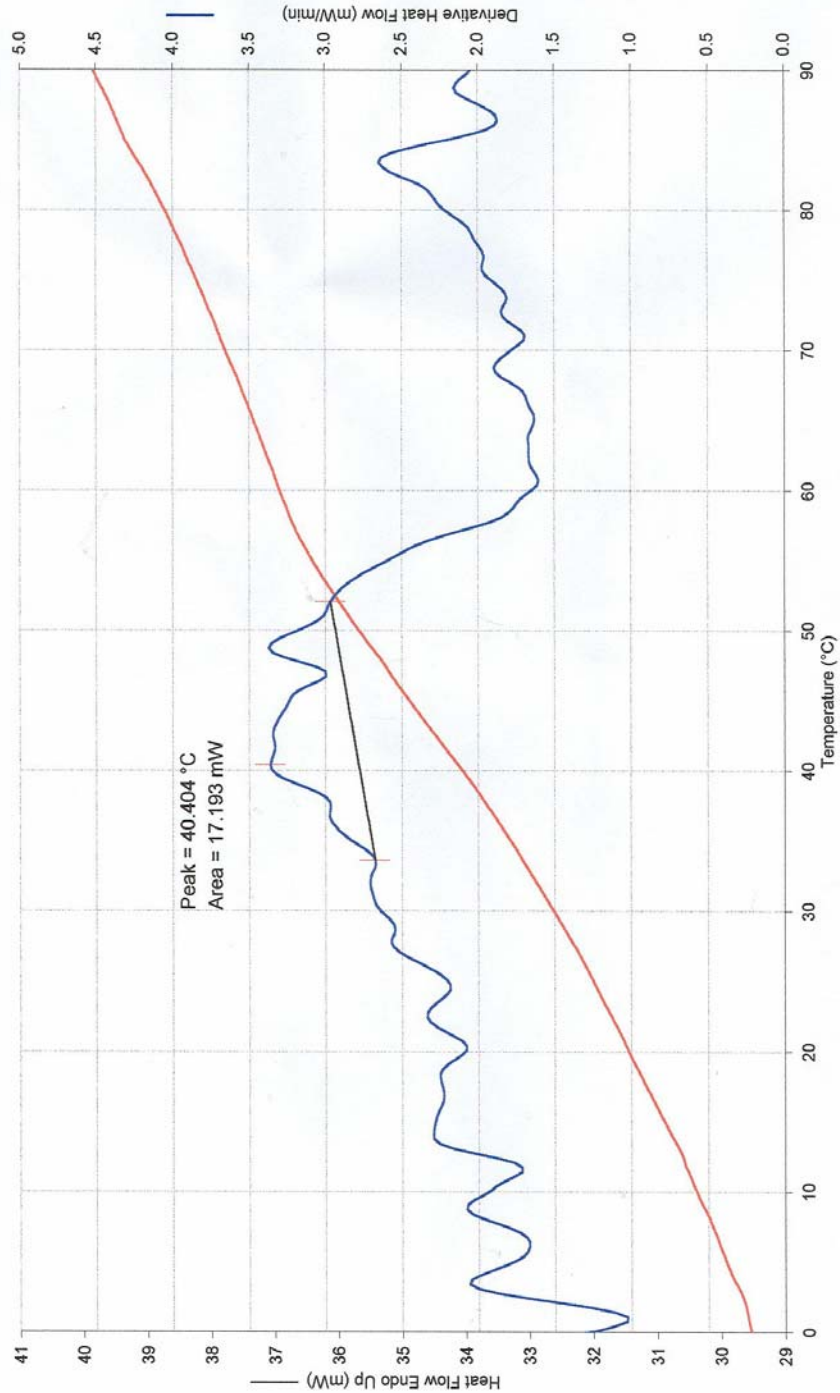


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...59725TG@060927111703.dsd
Operator ID: D. Anderoyk
Sample ID: NTSB S/O 59725
Sample Weight: 21,000 mg
Comment: STD 50/50 (1st Scan)



09/27/2006 12:44:51 PM

2) Hold for 15.0 min at 100.00°C

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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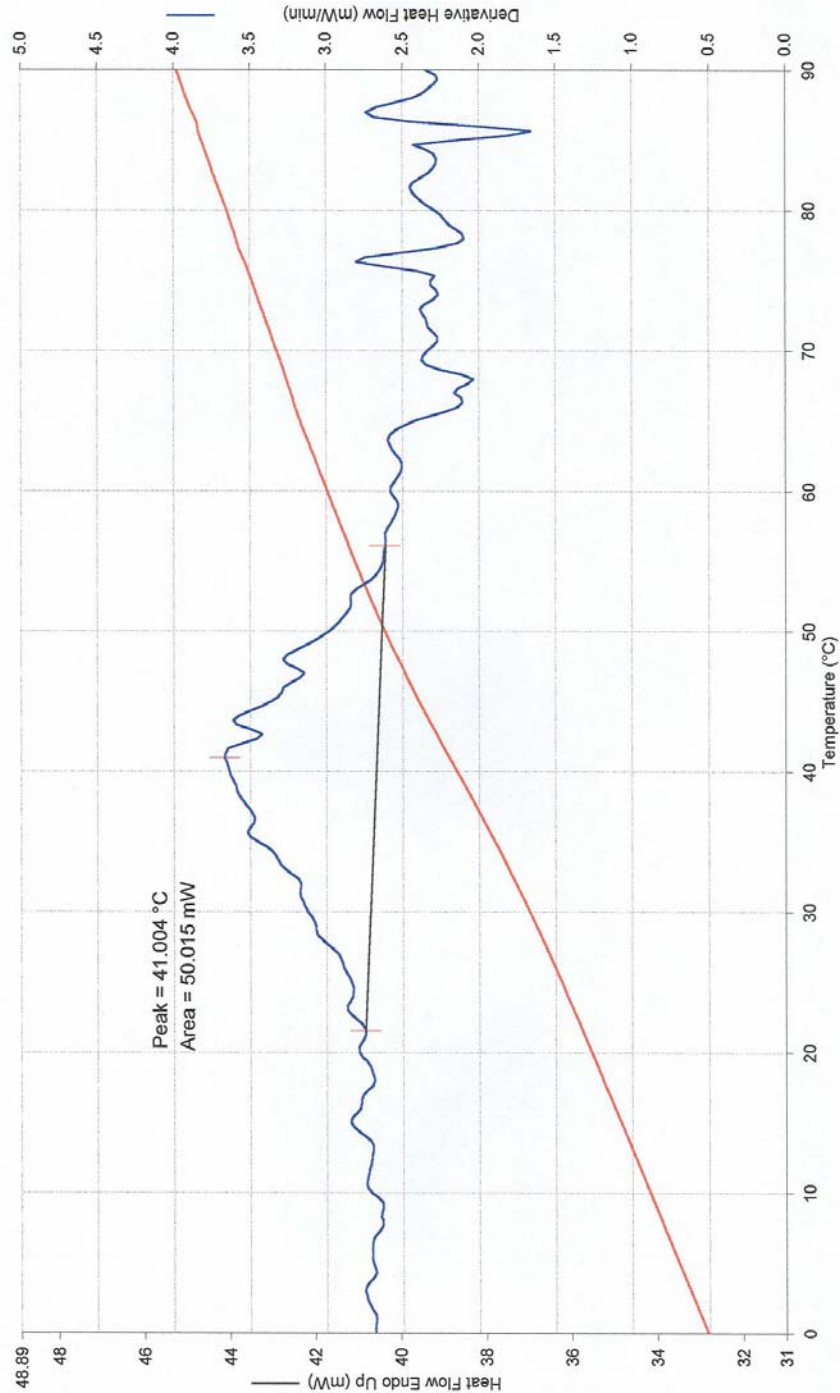


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...\59725TG@060927120153.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 21.000 mg
Comment: STD 50/50 (2nd Scan)



09/27/2006 12:42:38 PM

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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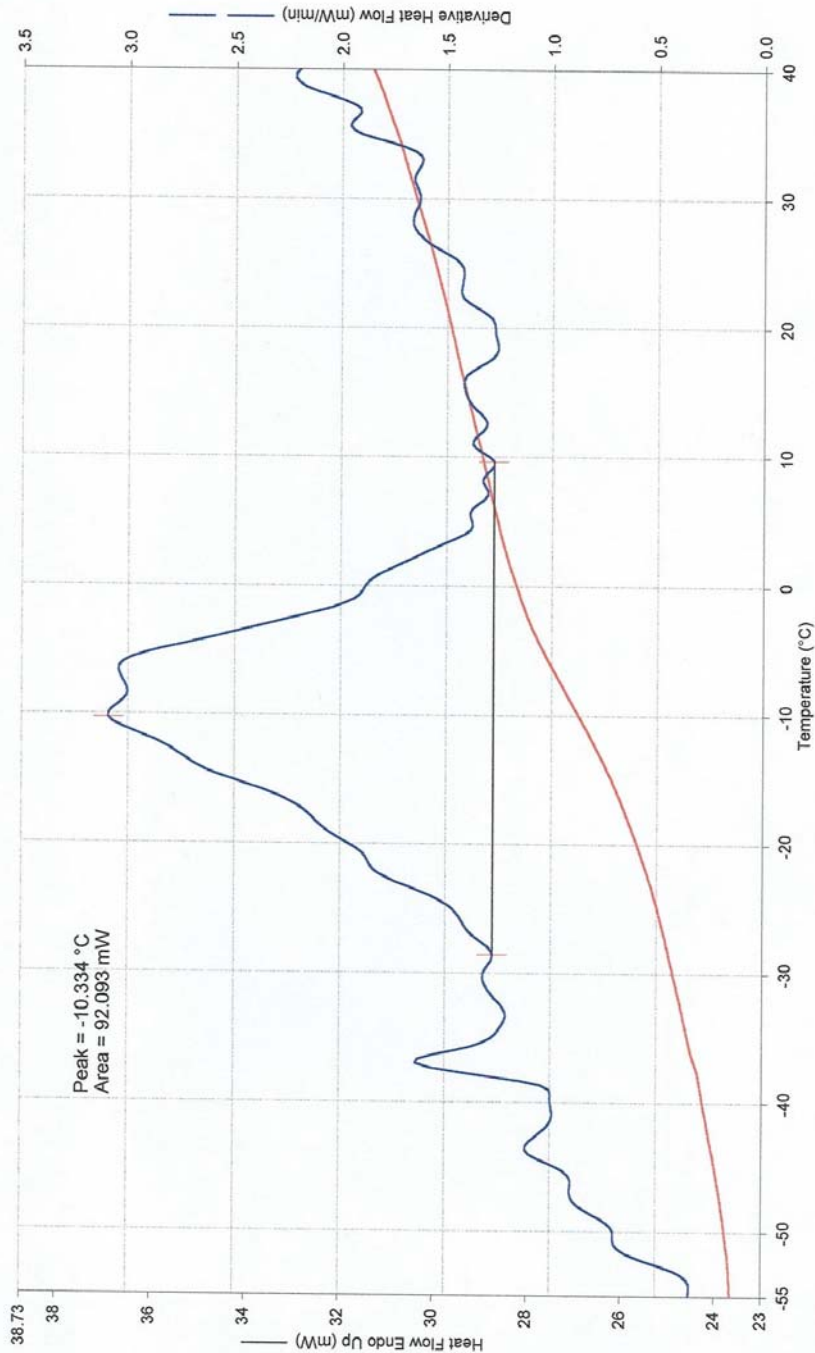


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...159725TG@060927103649.dsd
Operator ID: D. Anderoyk
Sample ID: NTSB S/O 59725
Sample Weight: 21.000 mg
Comment: STD 70/30 (1st Scan)



09/27/2006 12:47:59 PM

2) Hold for 15.0 min at 100.00 °C

1) Heat from -65.00 °C to 100.00 °C at 20.00 °C/min





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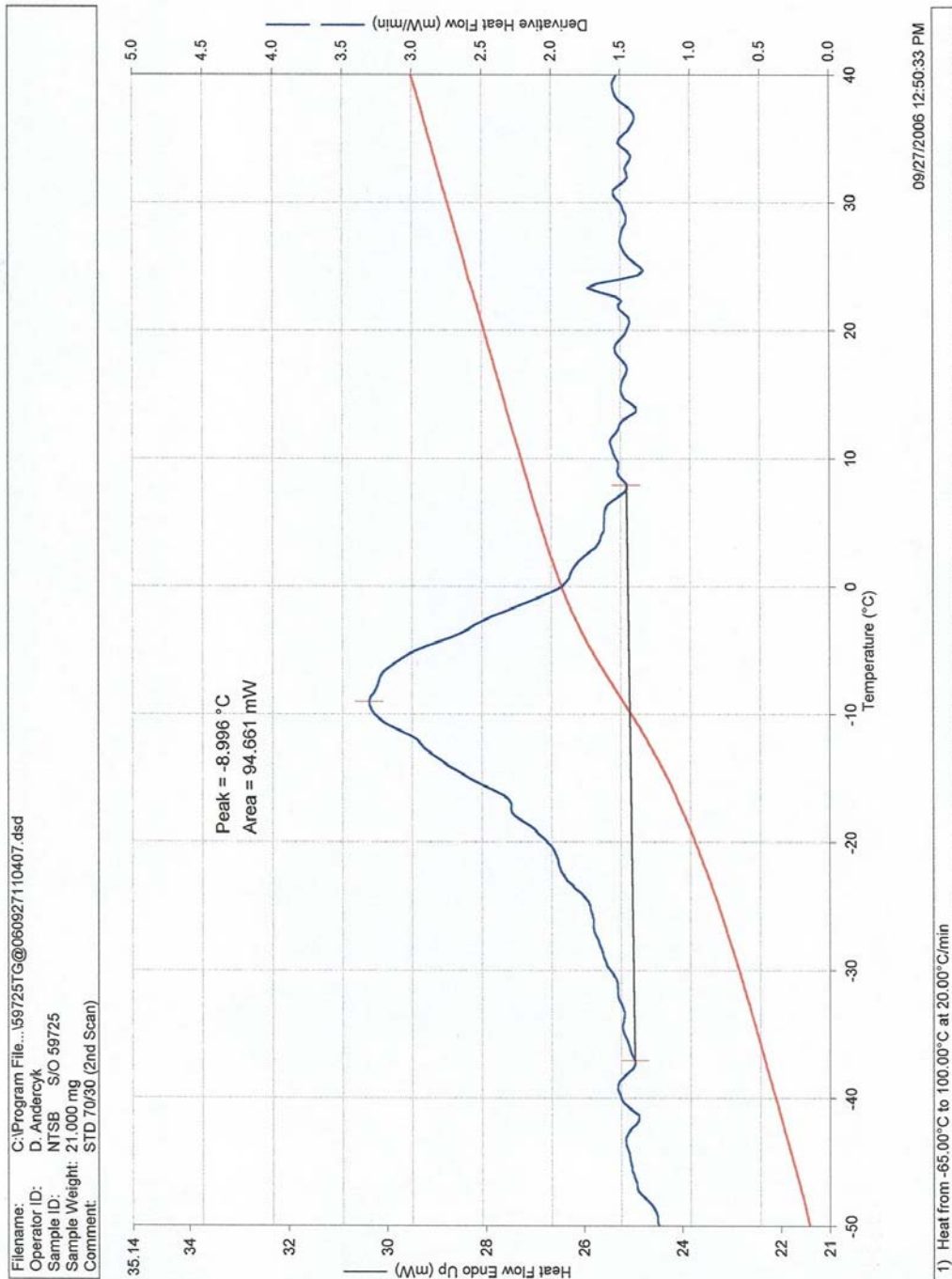
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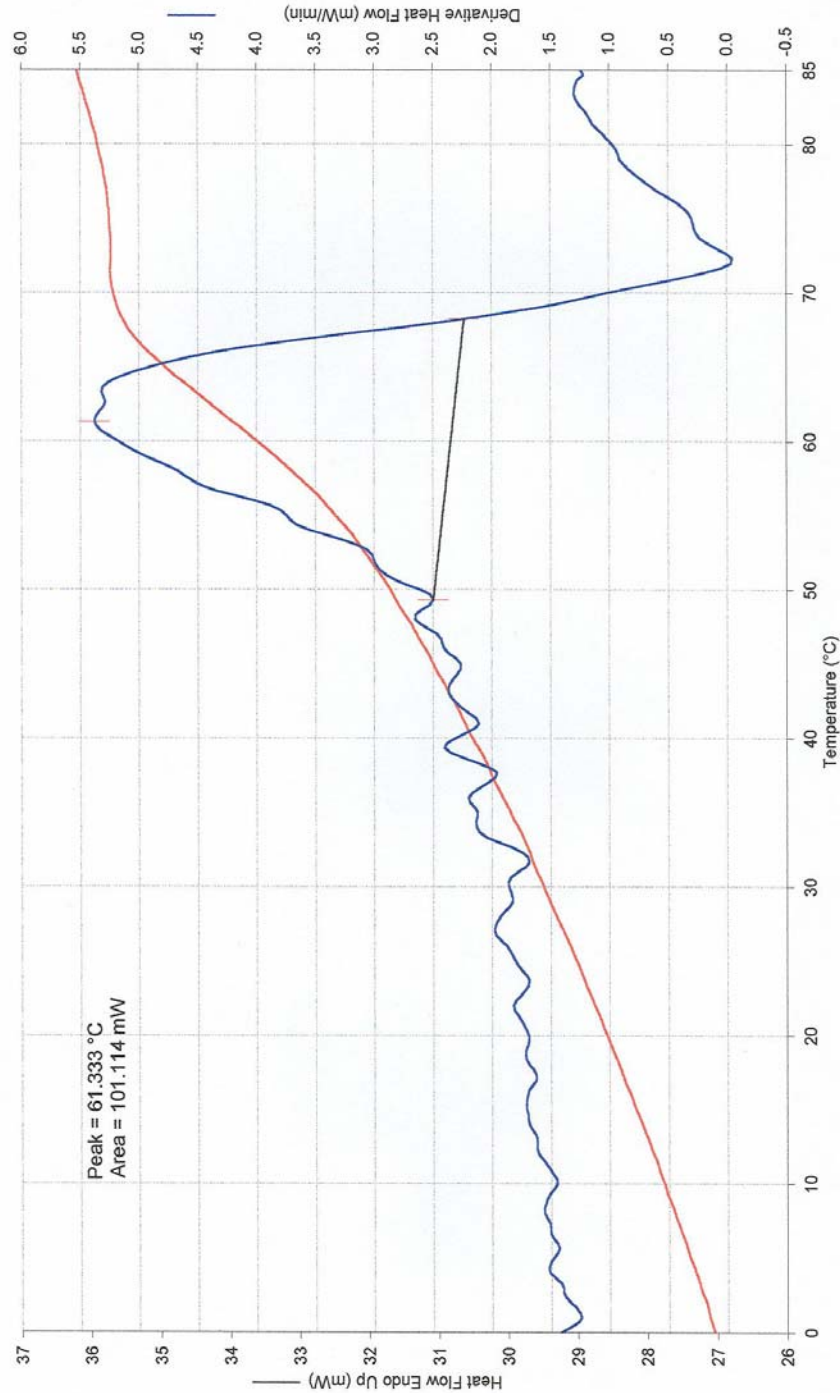


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Cert No. C2005-01504

Filename: C:\Program File... \59725TG@060927124642.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 21.100 mg
Comment: Fast 50/50 (1st Scan)



09/27/2006 1:11:44 PM

2) Hold for 15.0 min at 100.00°C

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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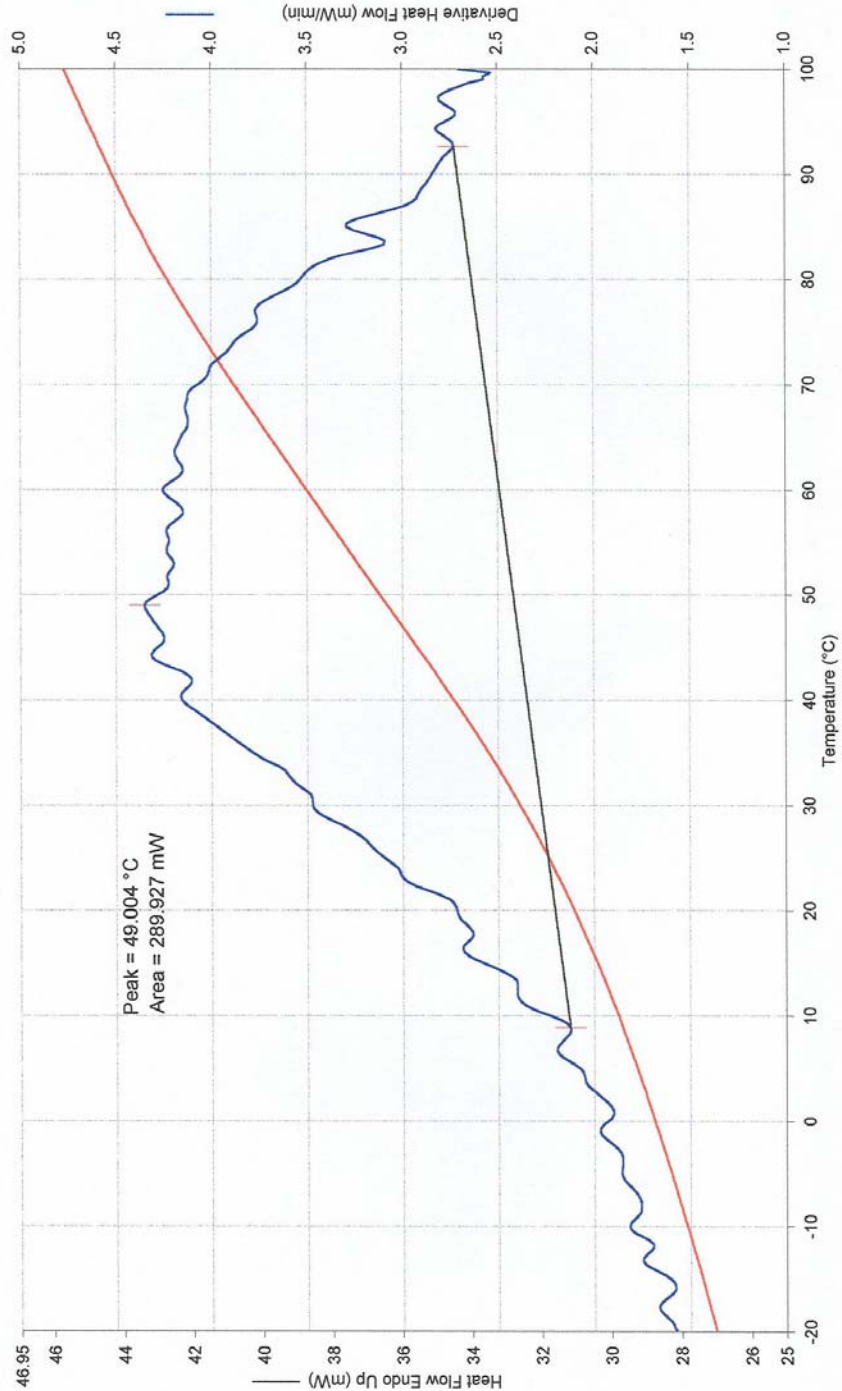


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program File...59725TG@060927131342.dsd
Operator ID: D. Andercyk
Sample ID: NTSB S/O 59725
Sample Weight: 21.100 mg
Comment: Fast 50/50 (2nd Scan)



09/27/2006 1:23:39 PM

1) Heat from -65.00°C to 100.00°C at 20.00°C/min





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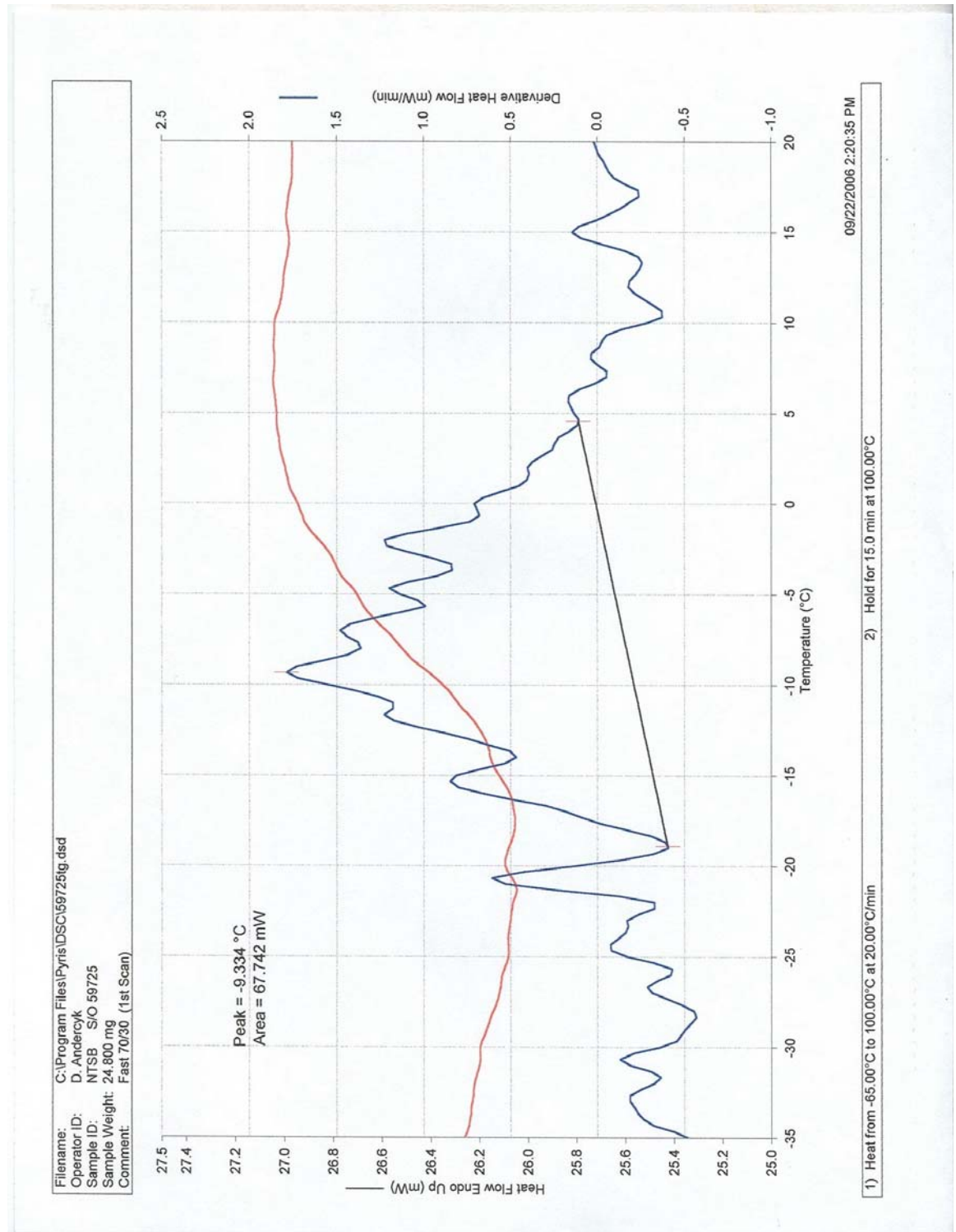
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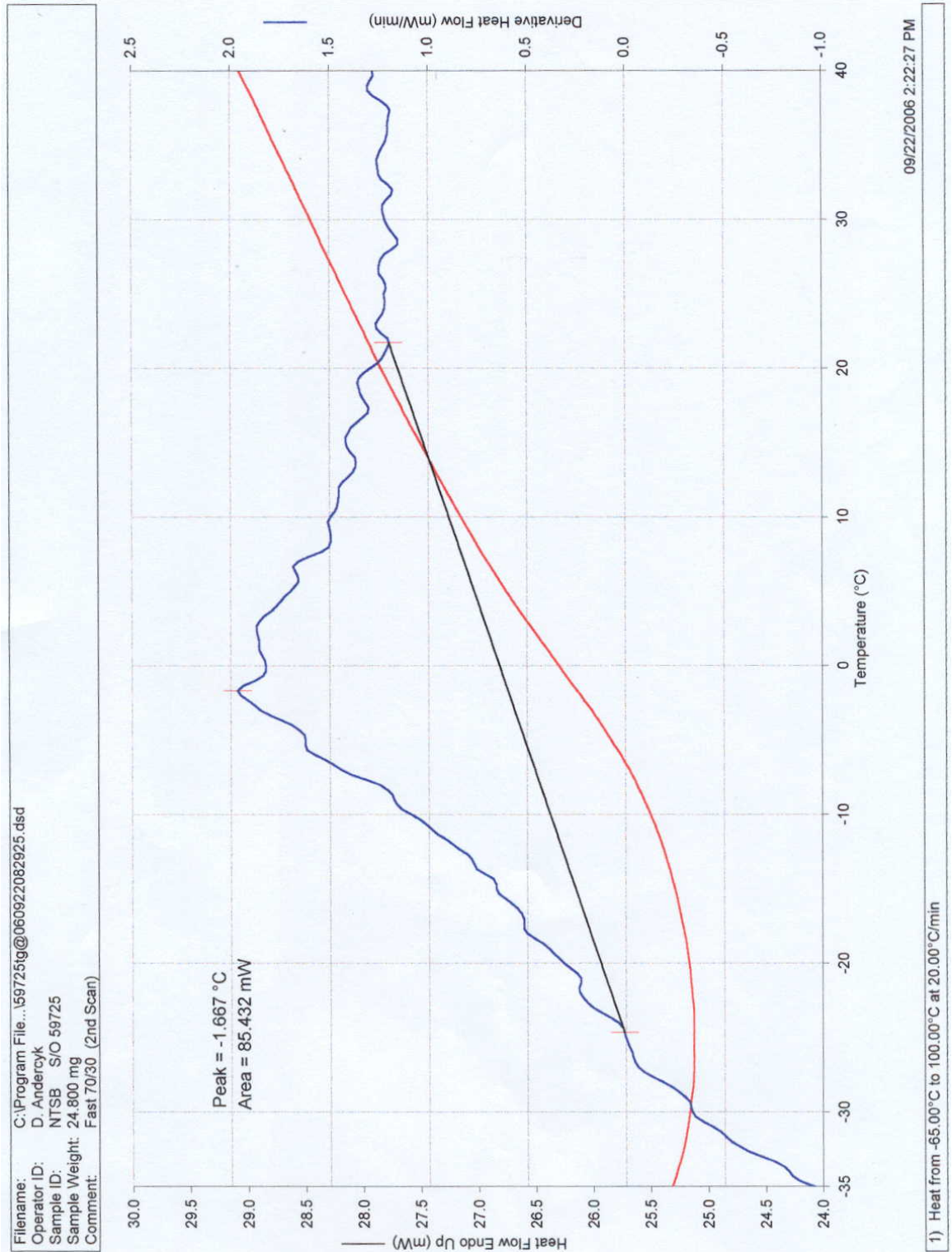
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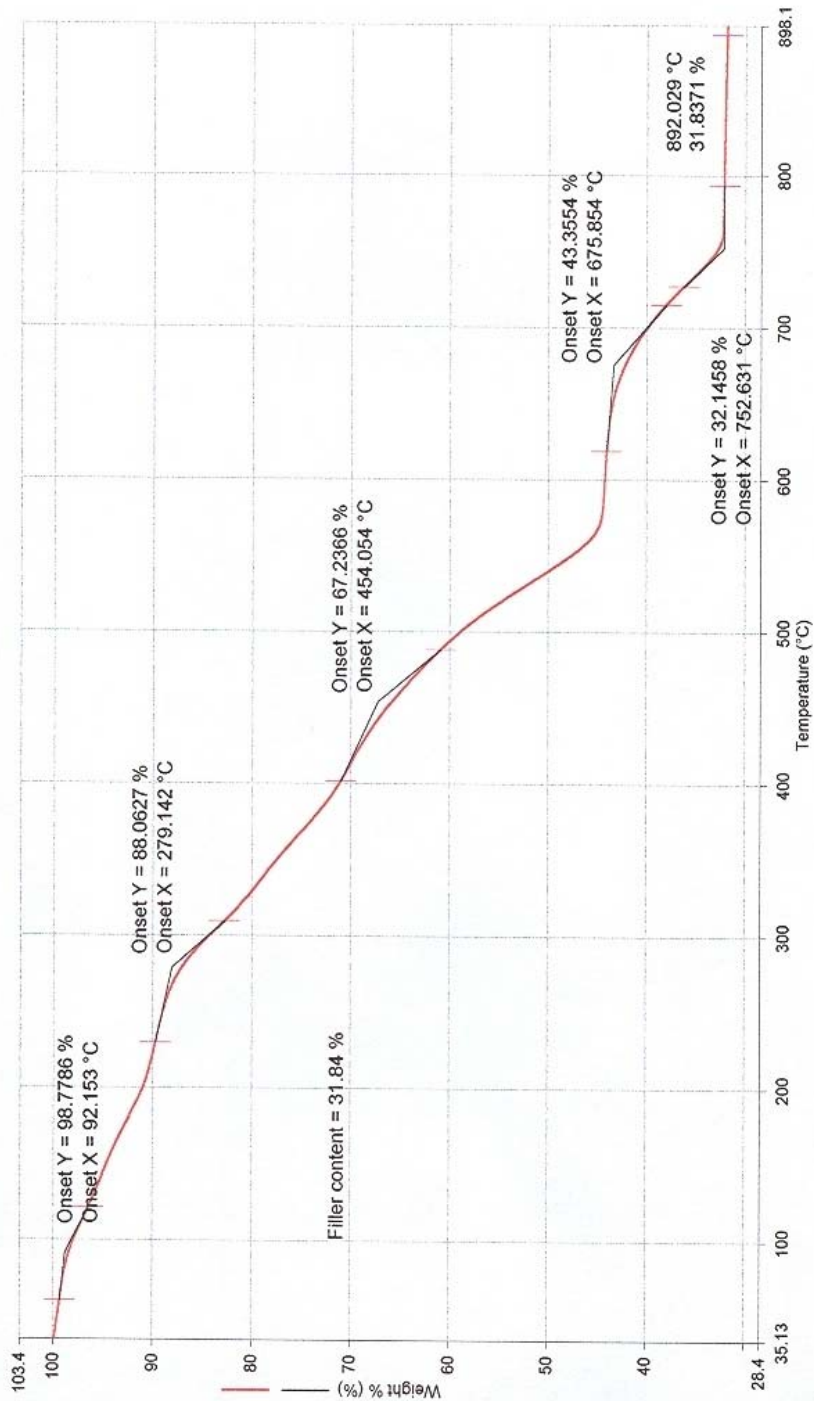


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program Files\Pyriss\...159725EB081M1.tgd
Operator ID: D. Phillips
Sample ID: NTSB S/O 59725
Sample Weight: 9.683 mg
Comment: HWY 06MA024
EB081 M1 2/3/4 NW



10/03/2006 8:03:11 AM

1) Heat from 30.00°C to 900.00°C at 10.00°C/min





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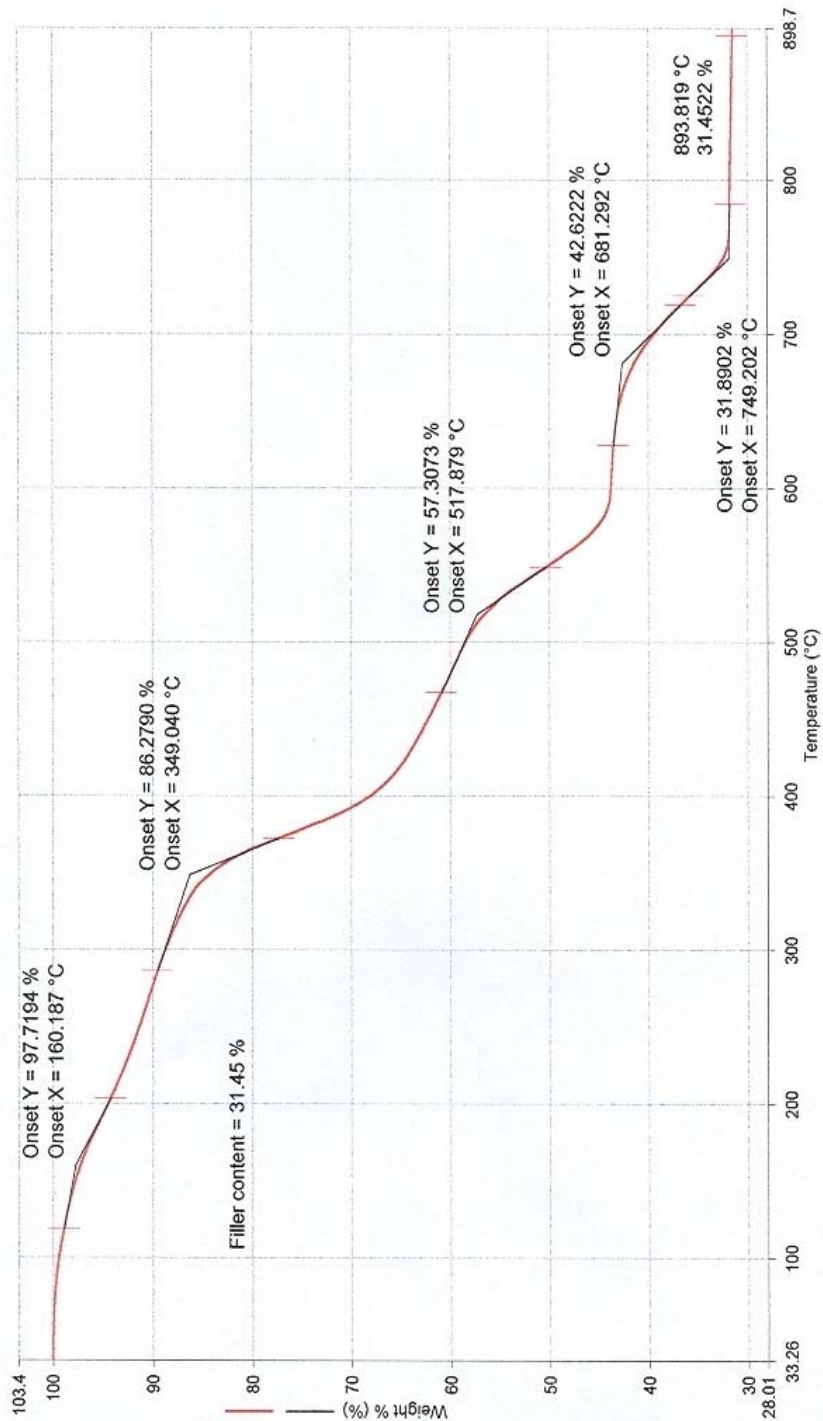


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program Files\Pyriss\159725\EB081M2.tgd
Operator ID: D. Phillips
Sample ID: NTSB S/O 59725
Sample Weight: 10.490 mg
Comment: HWY 06MA024
EB 081 M2 2/3/4 SE



10/03/2006 8:03:04 AM

1) Heat from 30.00°C to 900.00°C at 10.00°C/min





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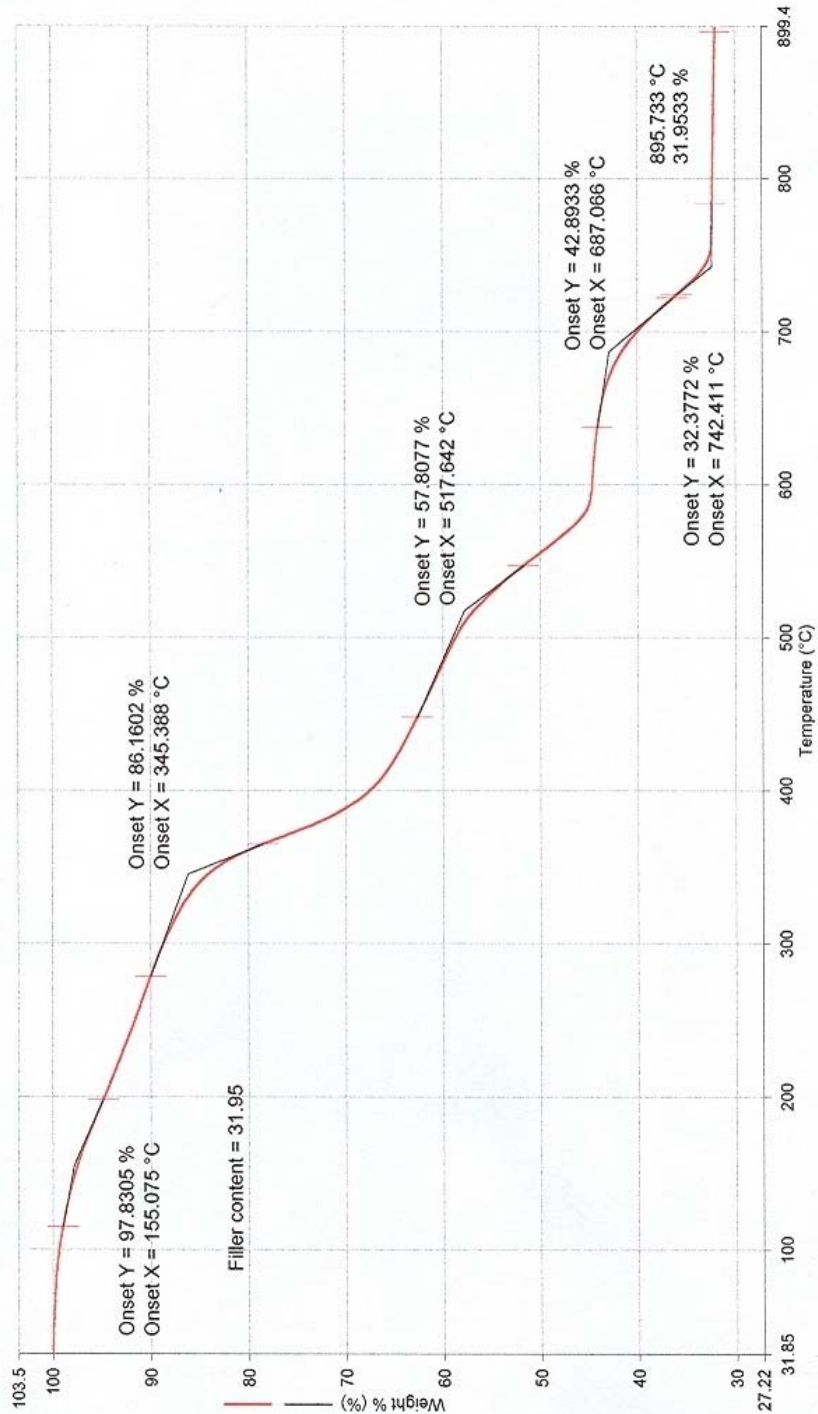


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program Files\Pyriss\TG...159725\WB082.tgd
Operator ID: D. Phillips
Sample ID: NTSB S/O 59725
Sample Weight: 9.288 mg
Comment: HWY 06MA024
WB 082 S 2 S



10/03/2006 8:02:54 AM

1) Heat from 30.00°C to 900.00°C at 10.00°C/min





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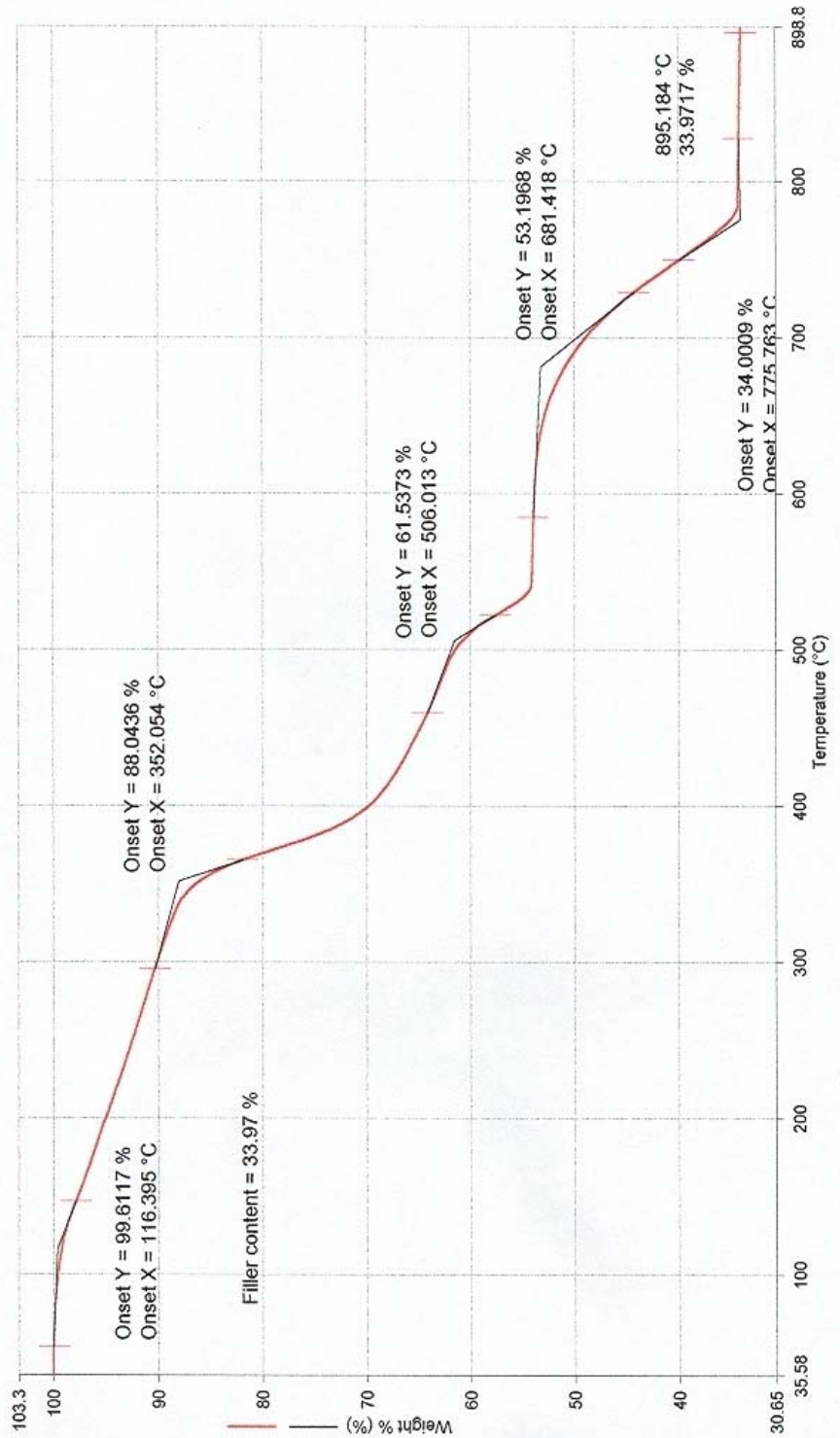


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program Files\Pyris\TG...159725std50.tgd
Operator ID: D. Phillips
Sample ID: NTSB S/O 59725
Sample Weight: 9.813 mg
Comment: HWY 06MA024
STD 50/50



10/03/2006 8:03:19 AM

1) Heat from 30.00°C to 900.00°C at 10.00°C/min





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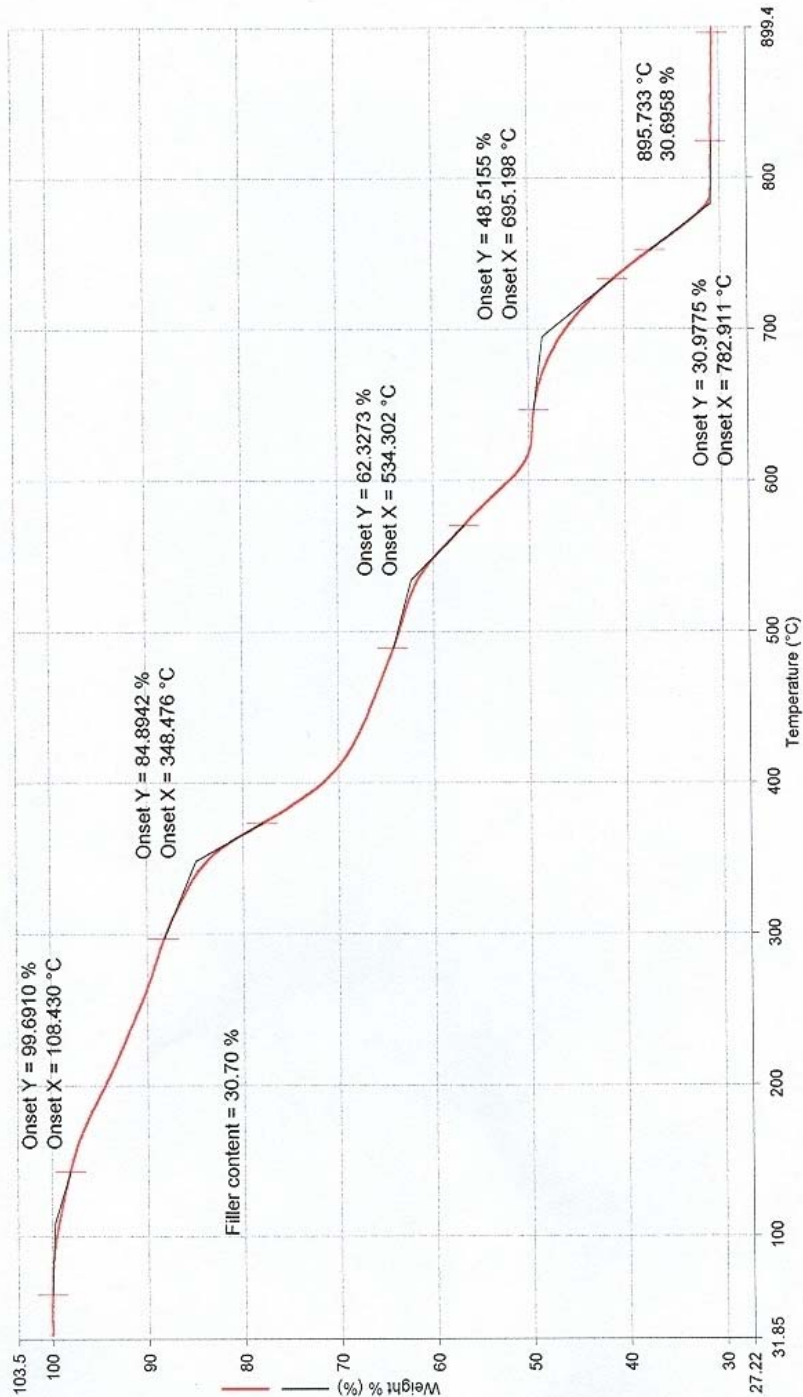


ISO/IEC 17025



Cert No. C2005-01504

Filename: C:\Program Files\PyratT...59725fast50.tgd
Operator ID: D. Phillips
Sample ID: NTSB S/O 59725
Sample Weight: 10.257 mg
Comment: HWY 06MA024
Fast 50/50



10/03/2006 8:02:38 AM

1) Heat from 30.00°C to 900.00°C at 10.00°C/min





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REGISTRARS, INC.

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TEST REPORT FOR:

NTSB
RE-30 Materials Laboratory Division
490 L'Enfant Plaza East, S.W.
Washington, DC 20594

Attn: Carl Schultheisz

DATE IN:

October 18, 2006

P/O #:

NTSB061012

SUBMISSION IDENTIFICATION:

Two (2) "known" and three (3) "unknown" epoxy samples were submitted for comparative analysis by dynamic headspace Gas Chromatograph Mass Spectroscopy (GCMS).

The three (3) "unknown" and two (2) "known" samples were serialized by the customer as listed:

"Unknown" Samples	"Known Samples"
• EB081 M1 2/3/4 NW	• Std 50/50 (HWY06MH024)
• EB081 M2 2/3/4 SE	• Fast 50/50 (HWY06MH024)
• WB082 S 2 S	

We offer our results and observations.

APPROVED:



Heather Beam
Manager Water Testing



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DYNAMIC HEADSPACE GAS CHROMATOGRAPHY MASS SPECTROSCOPY (GCMS)

TEST SPECIMENS:

Two (2) “known” epoxy samples, identified as:

“Known” Samples

- Std 50/50 (HWY06MH024)
- Fast 50/50 (HWY06MH024)

Three (3) “unknown” epoxy samples, identified as:

“Unknown” Samples

- EB081 M1 2/3/4 NW
- EB081 M2 2/3/4 SE
- WB082 S 2 S

SAMPLE PREPARATION:

“Known” Samples:

Hardened bulk samples were individually contained within a waxed paper laboratory cup (c.15 mL) and stored in plastic bags. Subsamples were taken from each bulk by drilling into the material with a clean 3/32 inch drill bit and collecting the shavings. Shavings were then compiled into a clean 2 mL autosampler/headspace vial, weighed, and sealed with a PTFE lined silicon septum (Table 1). Samples were exposed to 105° C for 24 hours within a laboratory oven. All samples were brought to room temperature before analysis.

“Unknown” Samples:

“Unknown” bulk samples arrived pulverized and needed no additional drilling or sub-sampling. These samples were prepared in the same manner and exposed to the same conditions as the “known” samples. See Table 1 for sample mass.

Table 1. Sample mass

• Std 50/50(HWY06MH024)	0.3286 g
• Fast 50/50(HWY06MH024)	0.3569 g
• EB081 M1 2/3/4 NW	0.4010g
• EB081 M2 2/3/4 SE	0.5754 g
• WB082 S 2 S	0.5368 g



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METHOD:

Equipment:

- Varian 4000 GCMS/MS
- Column: Varian FactorFour,™ 30m X 0.25mm VF-5ms, Df= 0.25µm (cat# CP8944).
- Injector: Varian type 1770
- Hamilton 1000µL Gastight® syringe (pt# 81330).
- Fisherbrand 2mL autosampler vials (cat# 0339116) with PTFE/silicone closures (cat# 0339117)
- Blue M Oven (pt# CC09SME/414217-1)

GCMS:

One (1) mL of headspace was withdrawn from each vial and transferred to the GCMS injector via gastight syringe. Eluting chromatographic compound peaks were identified by mass spectroscopy under set conditions (Table 2).

Table 2. GCMS Conditions

Injector:

Time	Split State	Split ratio
Initial	On	100
2.00	On	20

Column Flow: 1.2 mL/min

Column Oven:

Temp. (°C)	Rate (°C/min.)	Hold (min.)	Total (min.)
40	0	5	5
220	10	2	25
300	15	1	31.33
320	25	1.87	34

Mass Range:

EI Full Scan 25-450 m/z



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RESULTS:

“Known” Samples

Std 50/50 (HWY06MH024):

The “standard cure” chromatogram produced several significant peaks that are unique to this sample:

- Xylene at retention time (RT) 8.32 minutes.
- Peak cluster at RT from 14.09 to 16.00 minutes. This cluster consists primarily of naphthalenes and methylbenzenes.

See Chromatogram 1.

Fast 50/50 (HWY06MH024):

The chromatogram for the fast cure also expressed several peaks characteristic to this particular sample type:

- Methoxybenzene at RT 8.93 min.
- Phenol at RT 10.31 min.
- Benzyl alcohol at RT 11.43 min.

See Chromatogram 2.

Benzyl alcohol was present in both samples at a RT of 11.43 minutes (Chromatogram 1,2).

“Unknown” Samples

EB081 M1 2/3/4 NW

EB081 M2 2/3/4 SE

WB082 S 2 S

The Chromatograms in all three samples bear resemblance to the “known fast cure” sample chromatogram (Chromatogram 2) in that they all share three common peaks (Chromatograms 3,4,5):

- Methoxybenzene at RT: 8.9 min.
- Phenol at RT 10.3 min
- Benzyl alcohol at RT 11.4 min.

These sample chromatograms also similarly lack the naphthalenes/methylbenzenes peak cluster found in the “standard cure” chromatogram at RT 14.09- 16.00 minutes.



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Trace Laboratories-East certifies that the test equipment used complies with the calibration test purposes of ISO 10012-1, ANSI/NCSL Z540-1-1994, and MIL-STD-45662A and that the data contained in this report is accurate within the tolerance limitation of this equipment.

All test procedures detailed within this report are complete. The results in this report relate only to those items tested. If any additional information or clarification of this report is required, please contact us. This test report shall not be reproduced except in full, without the written approval of Trace Laboratories-East.

Thank you for selecting Trace Laboratories-East for your testing purposes.

PERFORMED BY:

[REDACTED]

Chad Miller
Scientist

Attachments: Five (6) Total:
Chromatogram 1
Chromatogram 2
Chromatogram 3
Chromatogram 4
Chromatogram 5
Chromatogram 6

SAMPLE DISPOSITION: Samples to be returned to customer.



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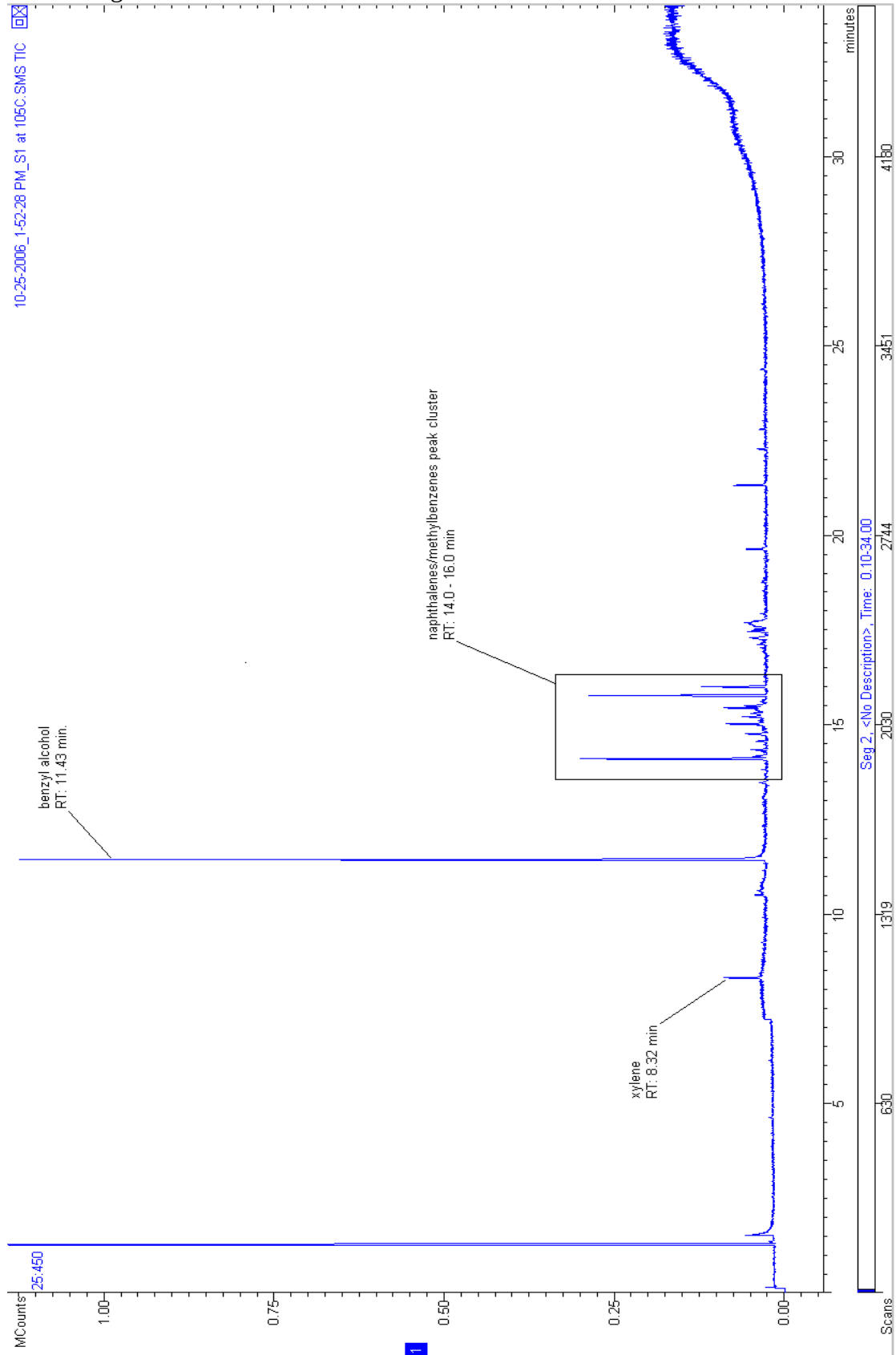


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Chromatogram 1. "Standard Cure"





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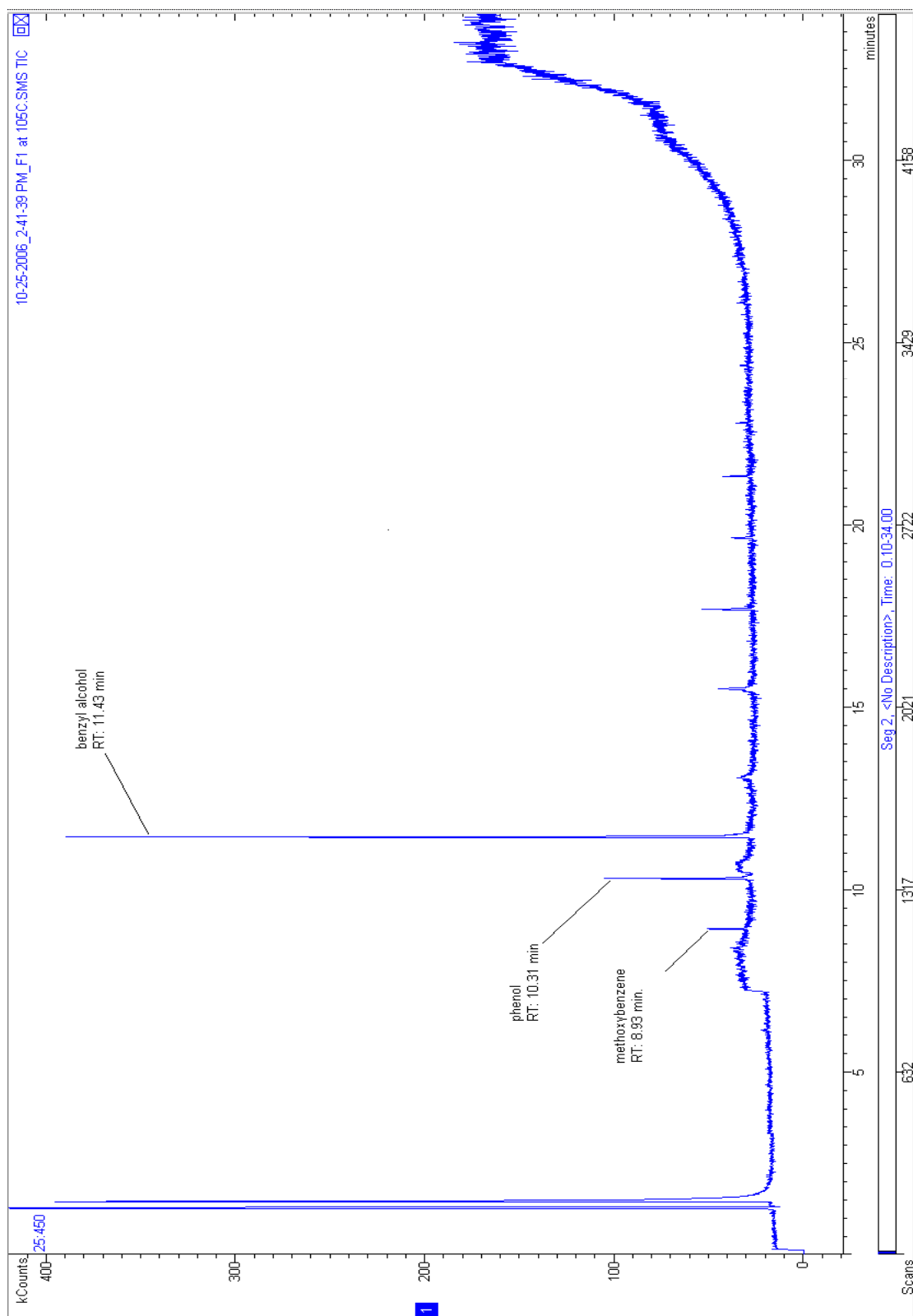


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Chromatogram 2 "Fast cure"





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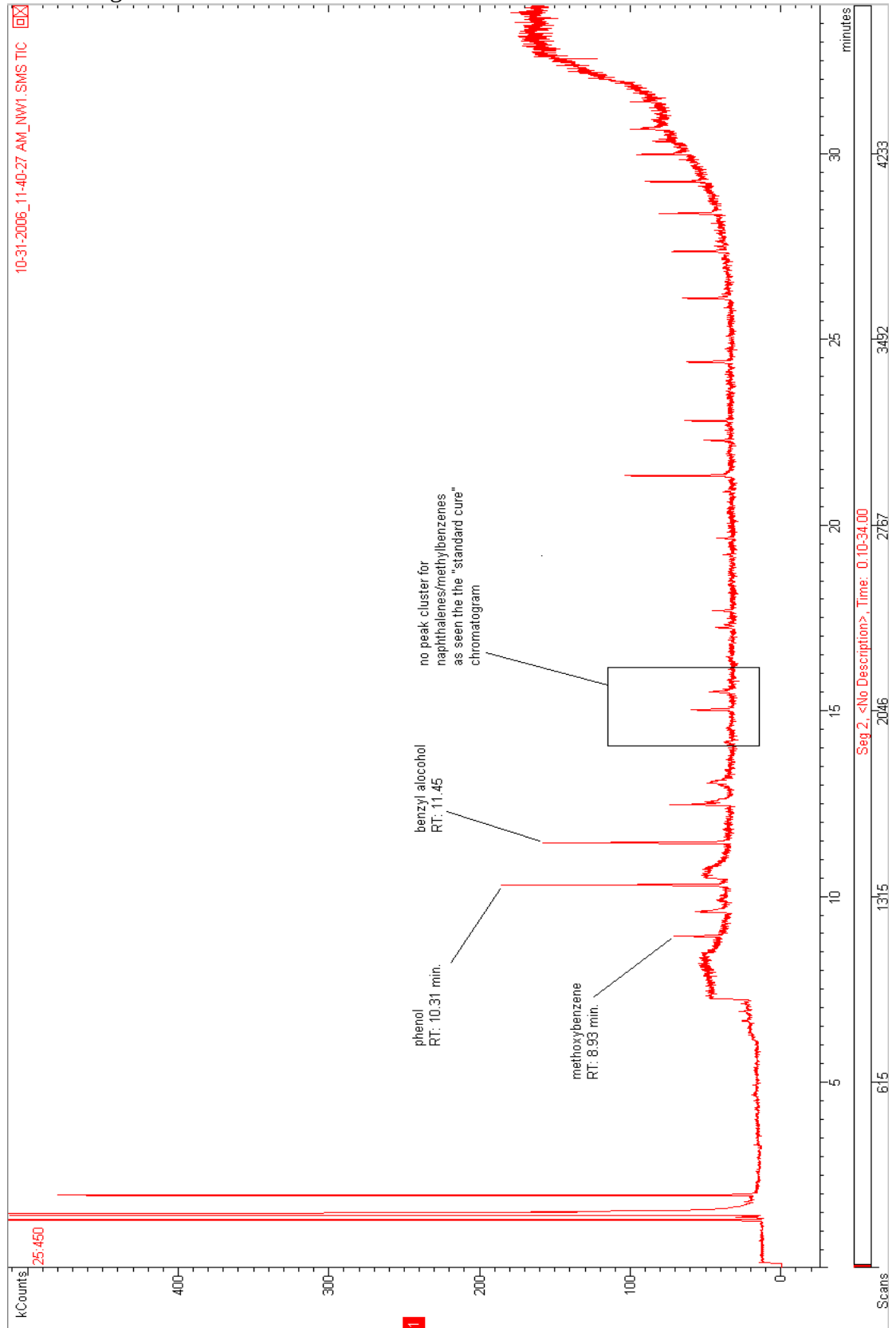
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Chromatogram 3 EB081 M1 2/3/4 NW "NW"





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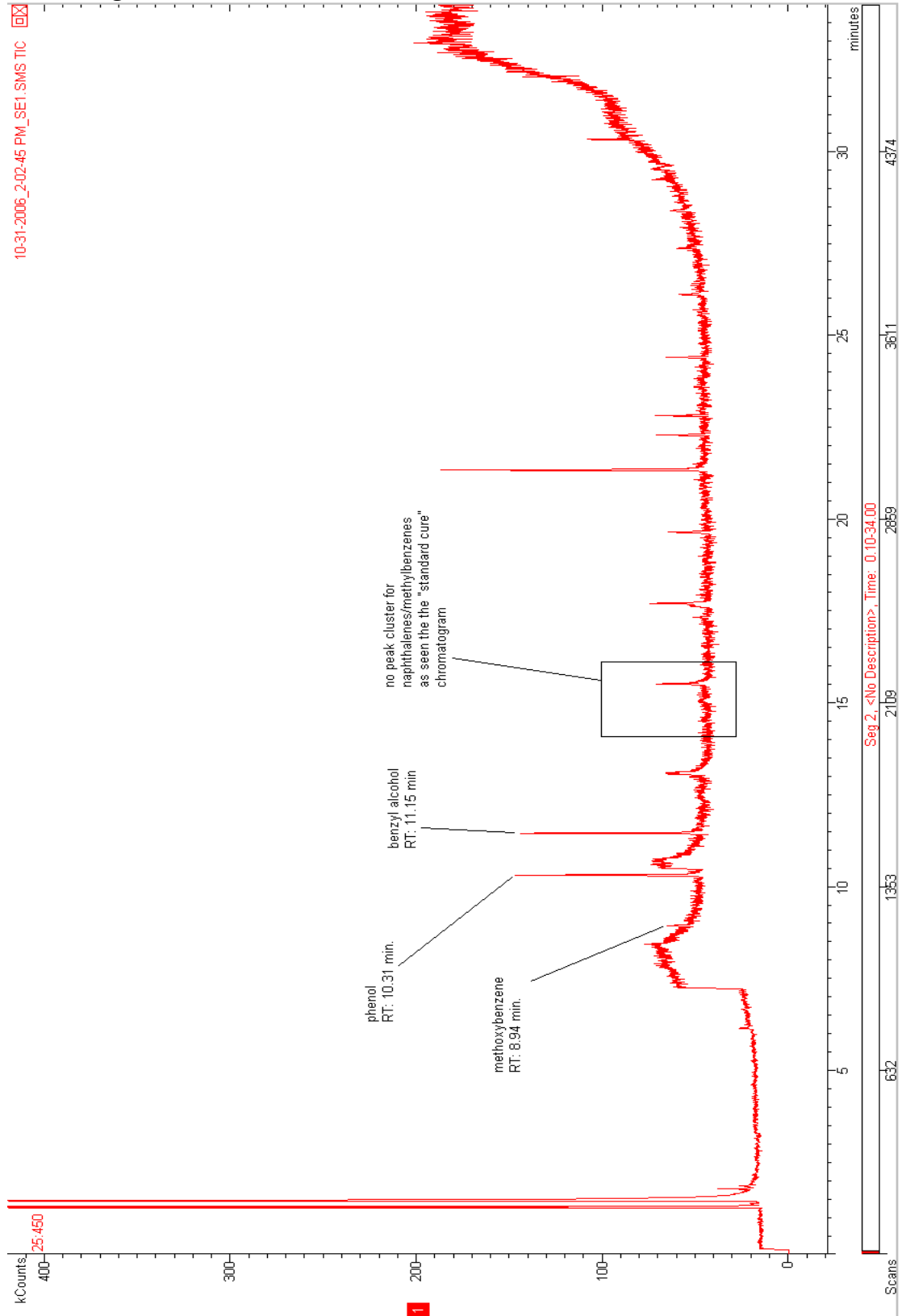


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Cert No. C2005-01504

Chromatogram 4 EB081 M2 2/3/4 SE "SE."





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Certificate No. TLE01A



Cert No(s) 1251.01, 1251.02,
1251.03

ISO 9001:2000



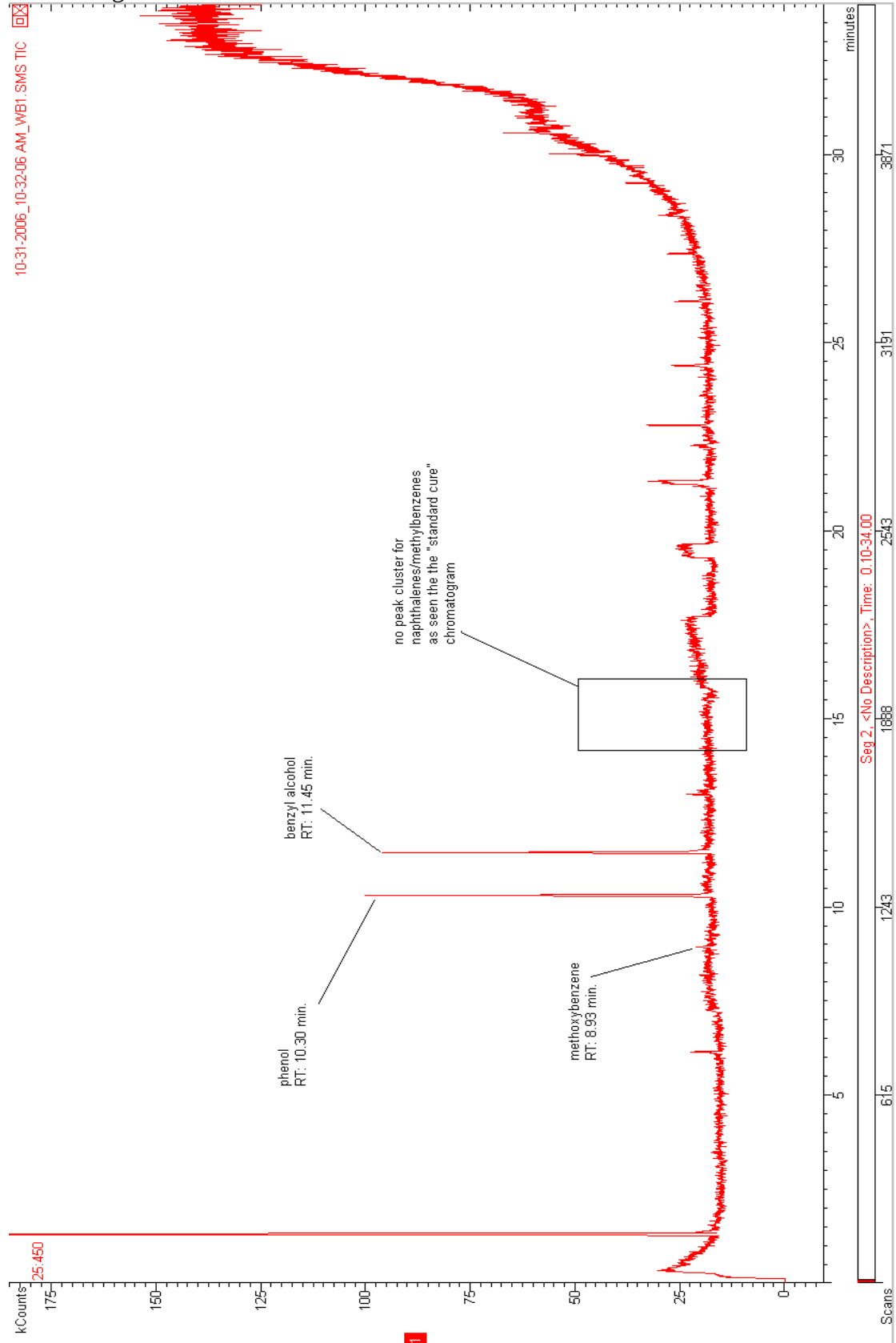
ISO/IEC 17025



PERRY JOHNSON
REGISTRARS, INC.

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Chromatogram 5 WB082 S 2 S "WB"





TRACE LABORATORIES-EAST

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Chromatogram 6 "Standard Cure," "fast Cure" overlay.

